

Sergei Giniatulin, Dconf 2026

Become a D user

From Zero to Hero



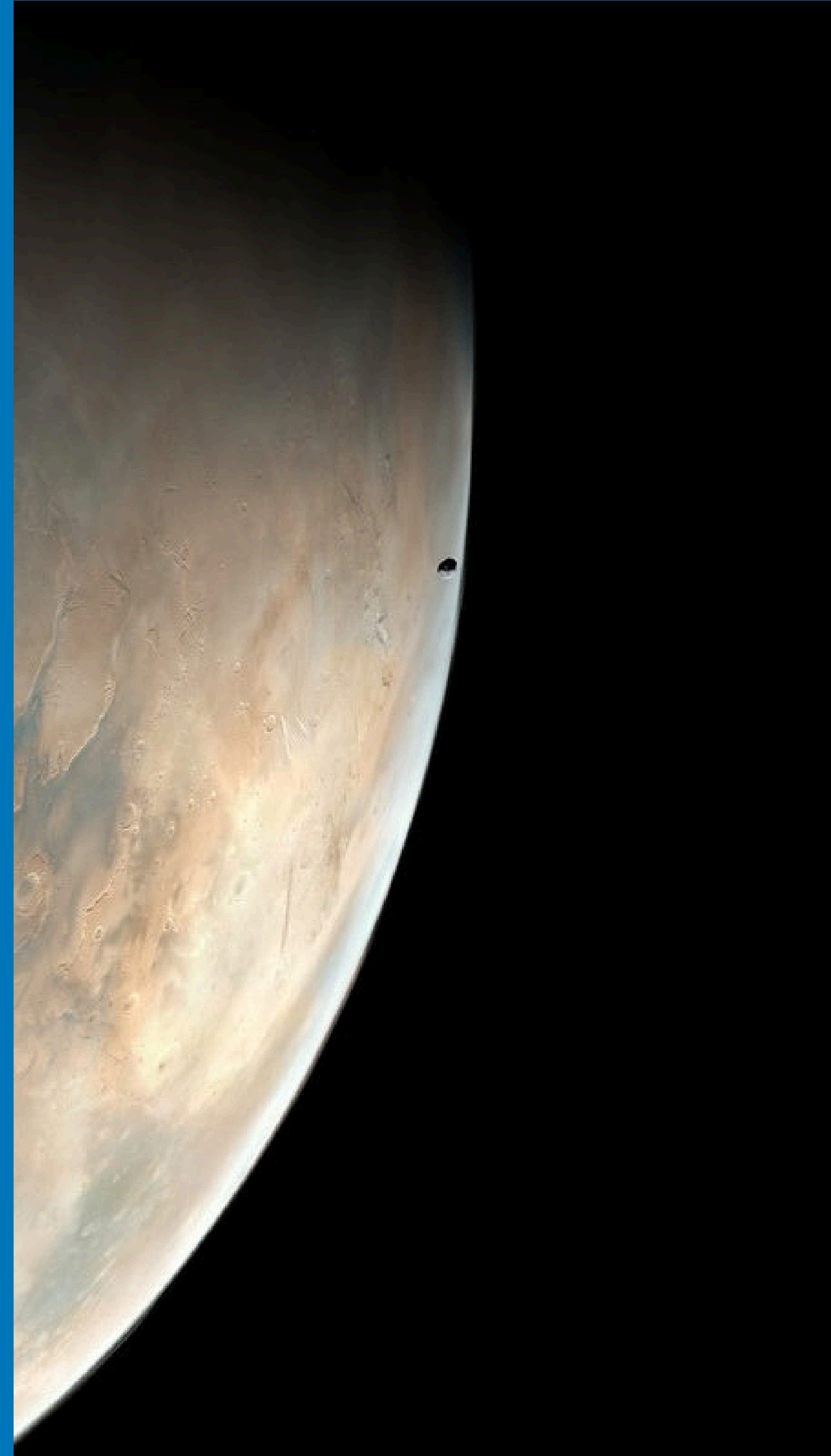
Agenda

- Beginning of the D story
- Path to grow
- Find your passion
- Learn from others



Flash Cards

Community appreciation



The Common Thread Behind It All

The Common Thread Behind It All

The butterfly effect

The Common Thread Behind It All

The butterfly effect Everything is intertwined

The Common Thread Behind It All

The butterfly effect Everything is intertwined

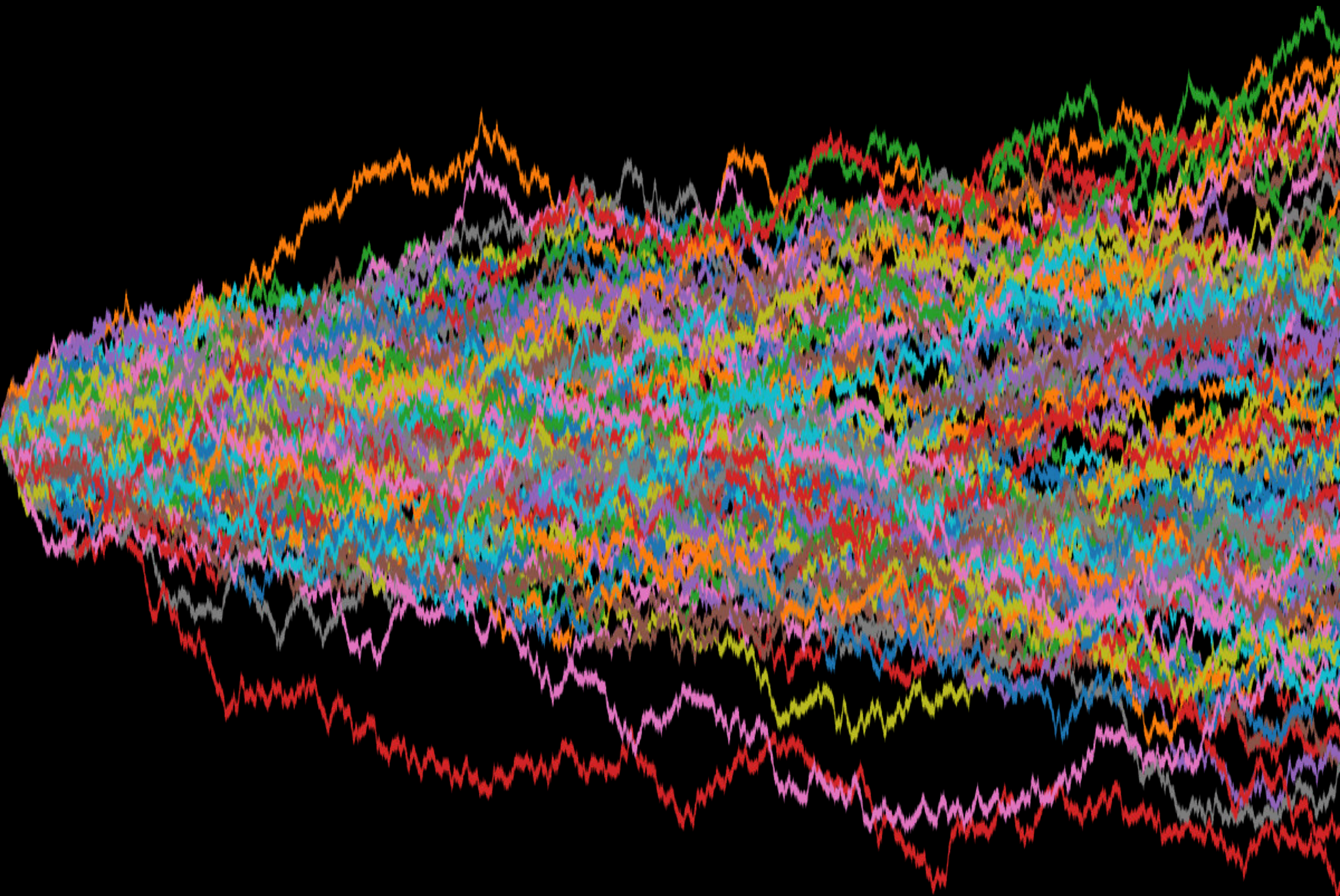
Life works in mysterious ways

The Common Thread Behind It All

The butterfly effect Everything is intertwined
You never can tell
Just give it a try
Life works in mysterious ways

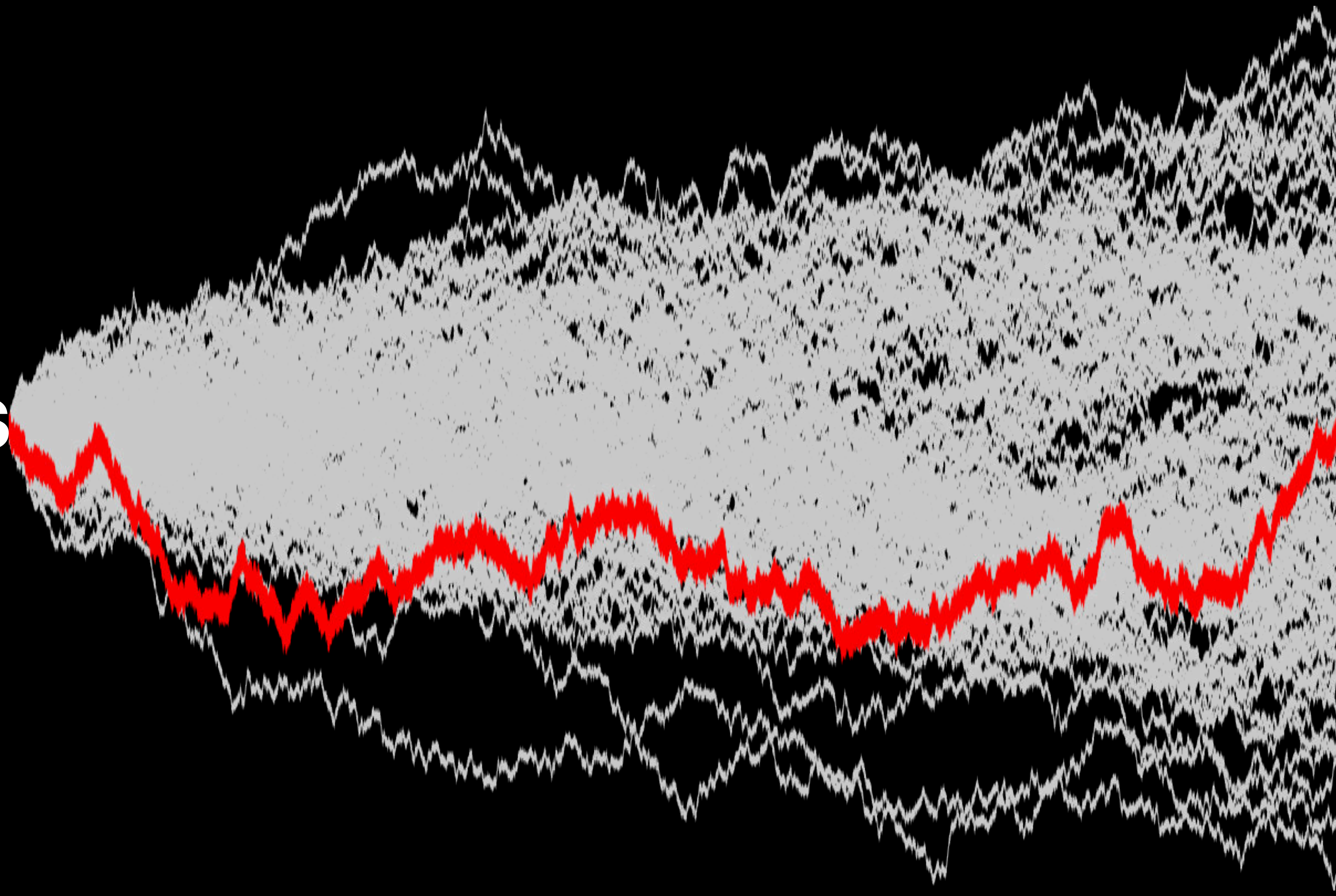
Different stories

About D language



Different stories

About D language



Background

For the context

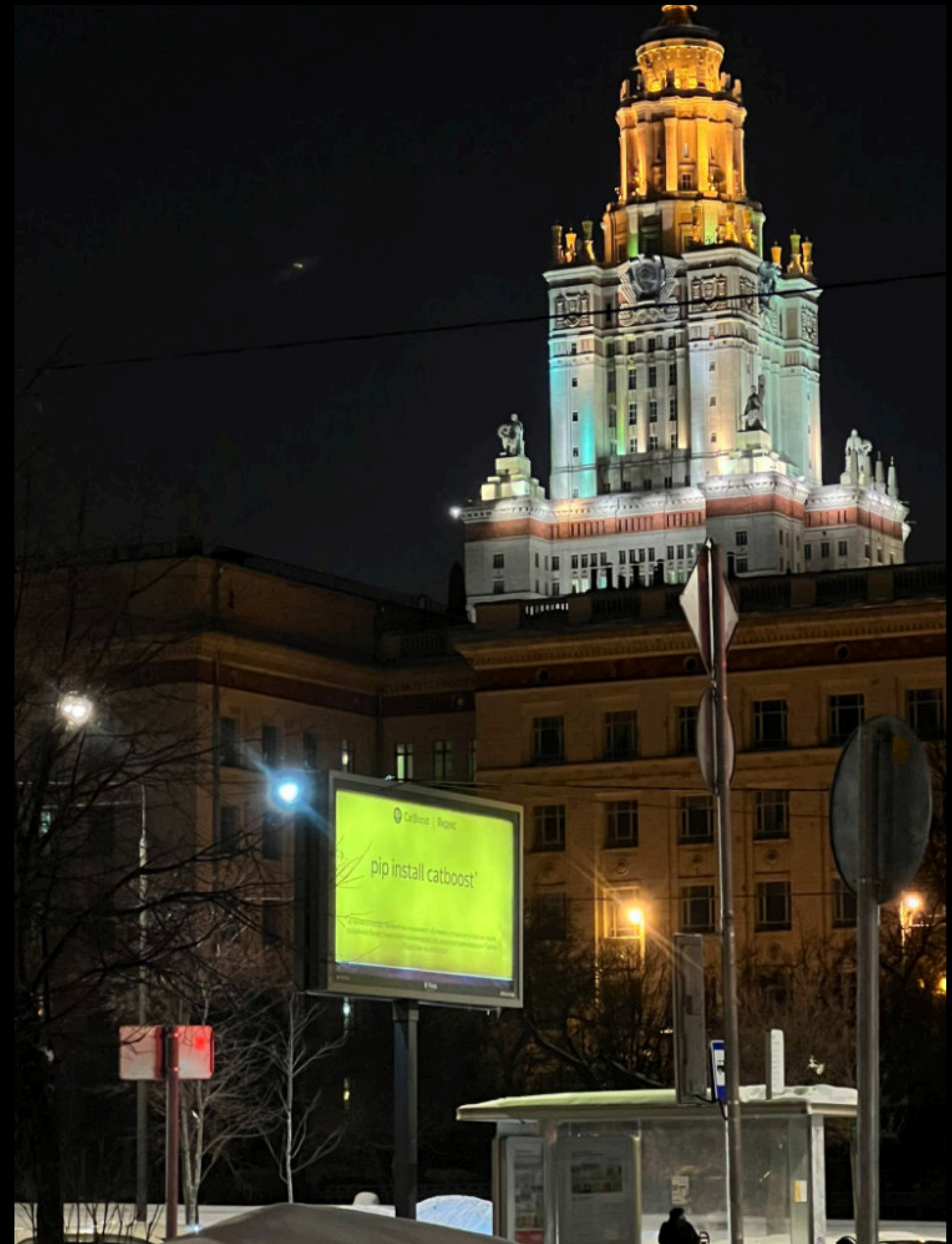
- MSU CMC (Math & CS)



Background

For the context

- MSU CMC (Math & CS)
- Pascal/MASM/C/C++/C#



Background

For the context

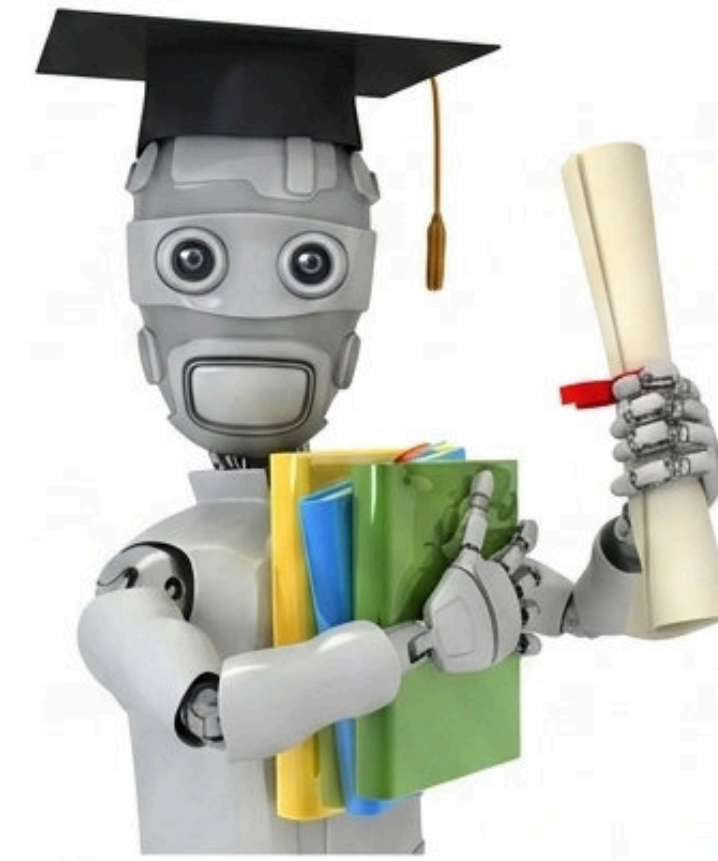
- MSU CMC (Math & CS)
- Pascal/MASM/C/C++/C#
- HPC & Fortran



Background

For the context

- MSU CMC (Math & CS)
- Pascal/MASM/C/C++/C#
- HPC & Fortran
- Machine Learning



Machine Learning

by Andrew Ng



Background

For the context

- MSU CMC (Math & CS)
- Pascal/MASM/C/C++/C#
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- Machine Learning

Data Analysis Tools



Discovery



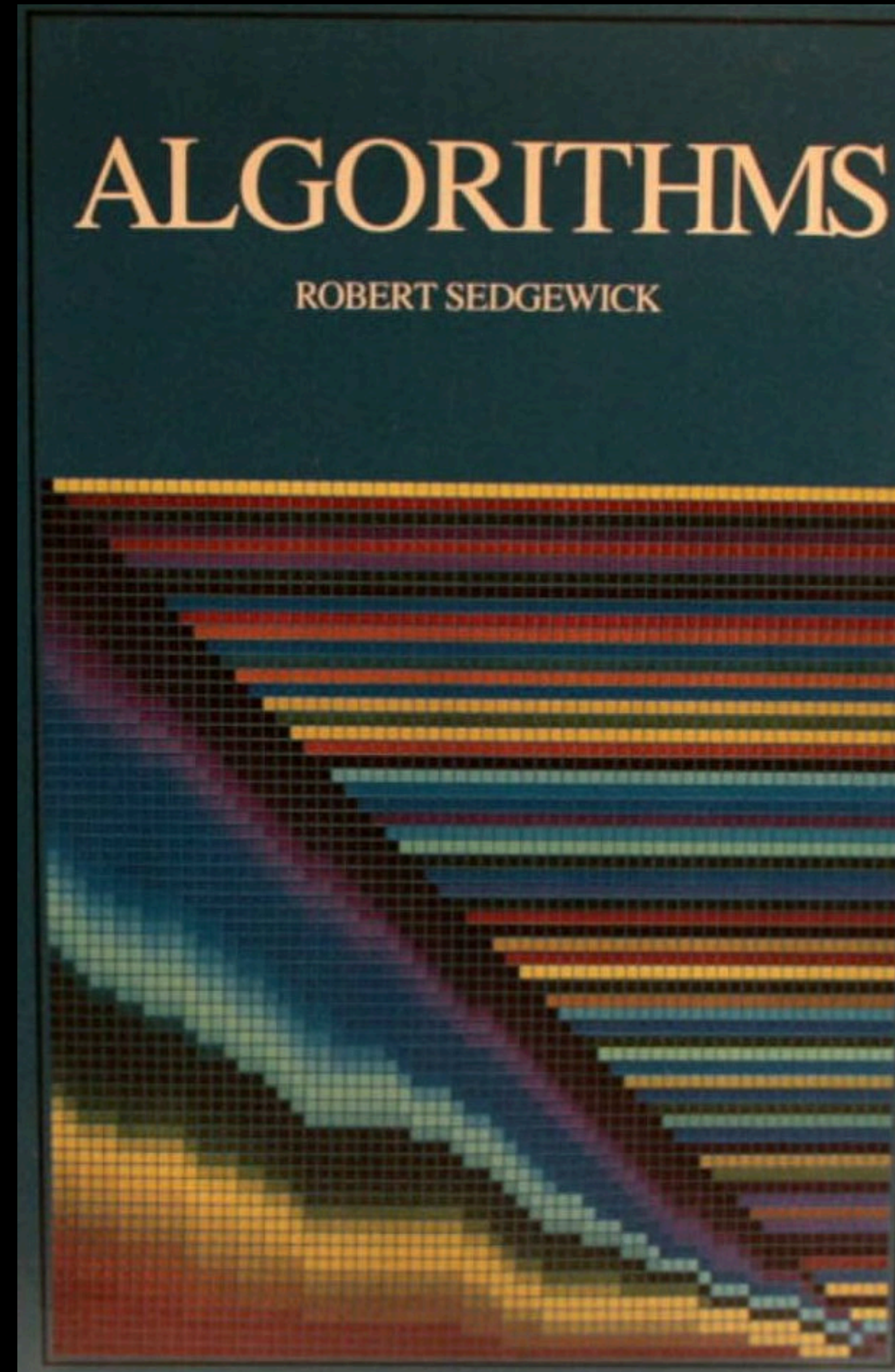
Discovery

- 3 semester educational program



Discovery

- 3 semester educational program
- Algorithms & Data Structures



Discovery

- 3 semester educational program
- Algorithms & Data Structures
- 2 main languages



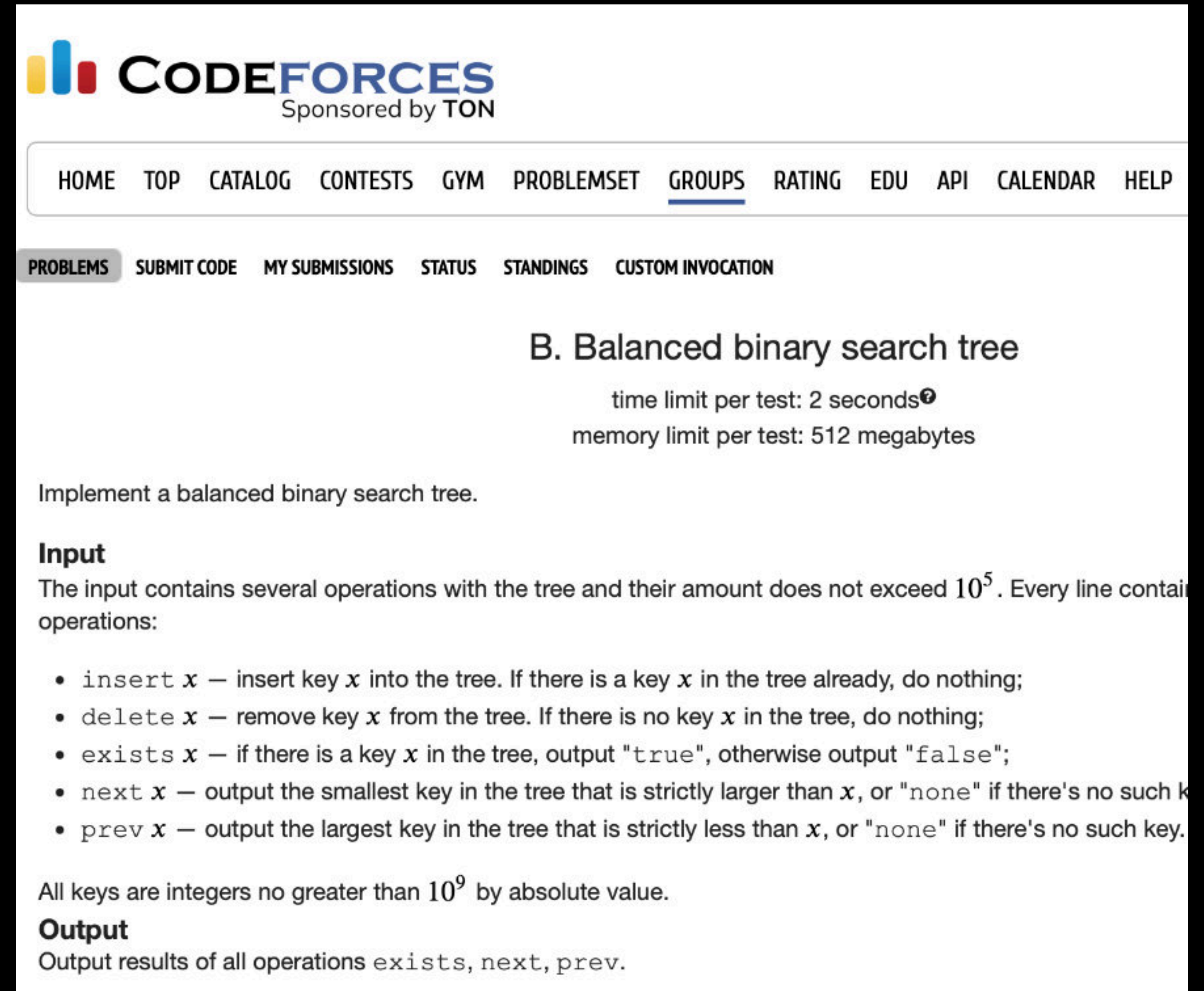
Discovery

- 3 semester educational program

- Algorithms & Data Structures

- 2 main languages

- Codeforces



The screenshot shows the Codeforces website interface. At the top, the Codeforces logo is displayed with the text 'Sponsored by TON'. Below the logo is a navigation bar with links: HOME, TOP, CATALOG, CONTESTS, GYM, PROBLEMSET, GROUPS (highlighted), RATING, EDU, API, CALENDAR, and HELP. Below this is a secondary navigation bar with links: PROBLEMS (highlighted), SUBMIT CODE, MY SUBMISSIONS, STATUS, STANDINGS, and CUSTOM INVOCATION. The main content area displays the problem title 'B. Balanced binary search tree' in bold. Below the title, the time limit is '2 seconds' and the memory limit is '512 megabytes'. The problem description states: 'Implement a balanced binary search tree.' The 'Input' section explains that the input contains several operations with the tree and their amount does not exceed 10^5 . Every line contains one operation:

- `insert x` — insert key x into the tree. If there is a key x in the tree already, do nothing;
- `delete x` — remove key x from the tree. If there is no key x in the tree, do nothing;
- `exists x` — if there is a key x in the tree, output "true", otherwise output "false";
- `next x` — output the smallest key in the tree that is strictly larger than x , or "none" if there's no such key;
- `prev x` — output the largest key in the tree that is strictly less than x , or "none" if there's no such key.

 The 'Output' section states: 'Output results of all operations exists, next, prev.'

Discovery

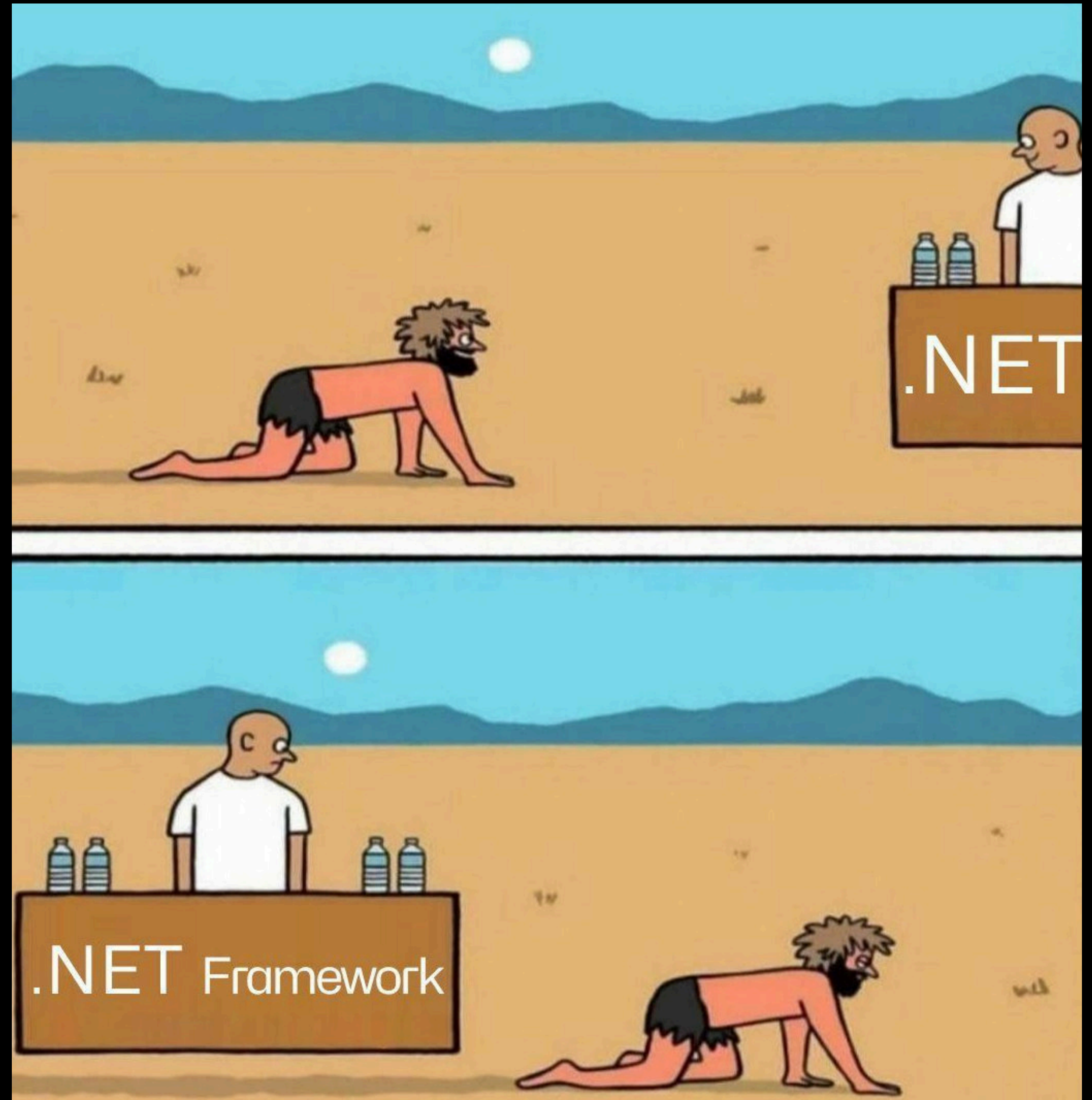
- 3 semester educational program
- Algorithms & Data Structures
- 2 main languages
- Codeforces
- But one day..

Slowest things on earth:



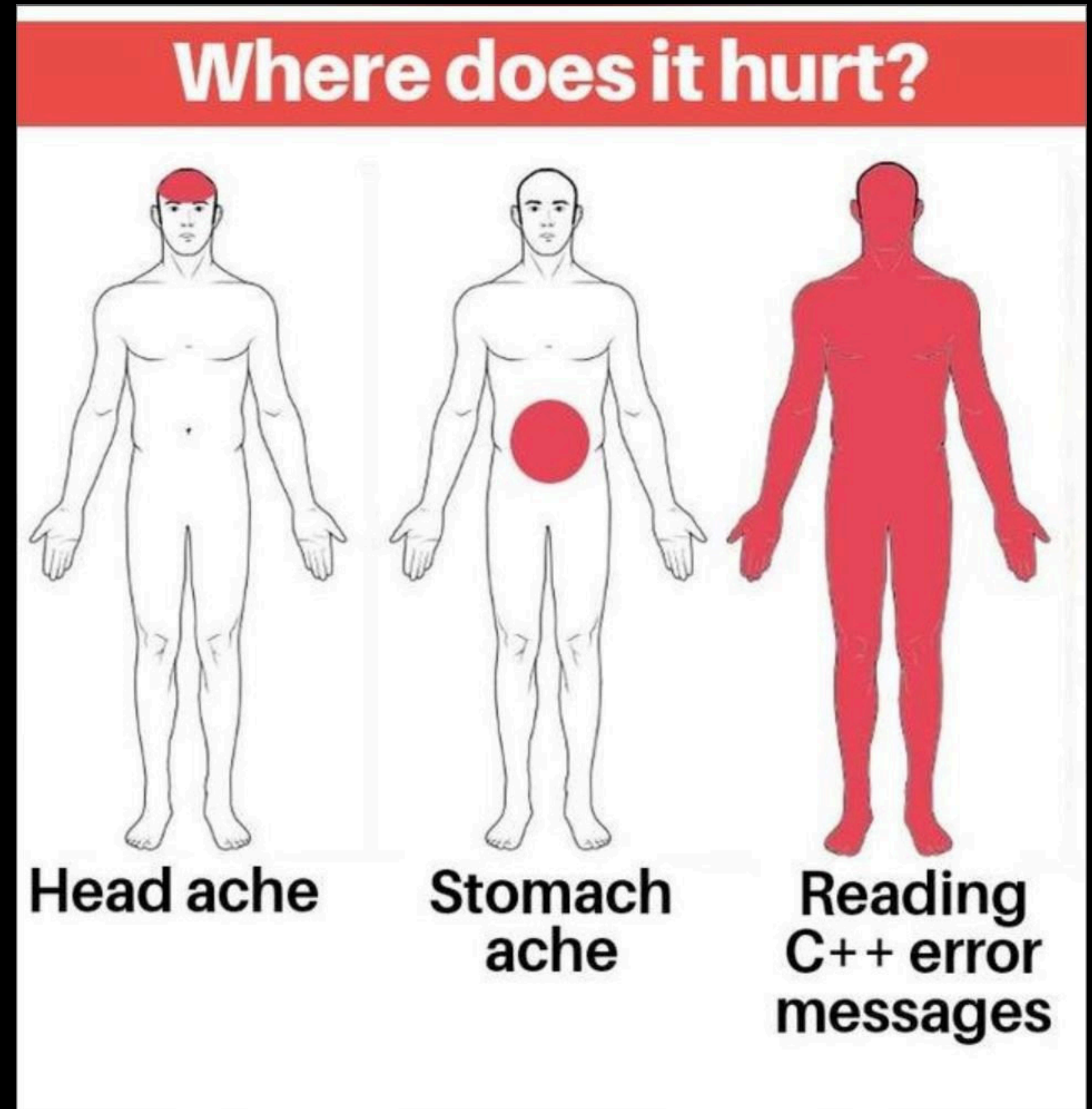
Discovery

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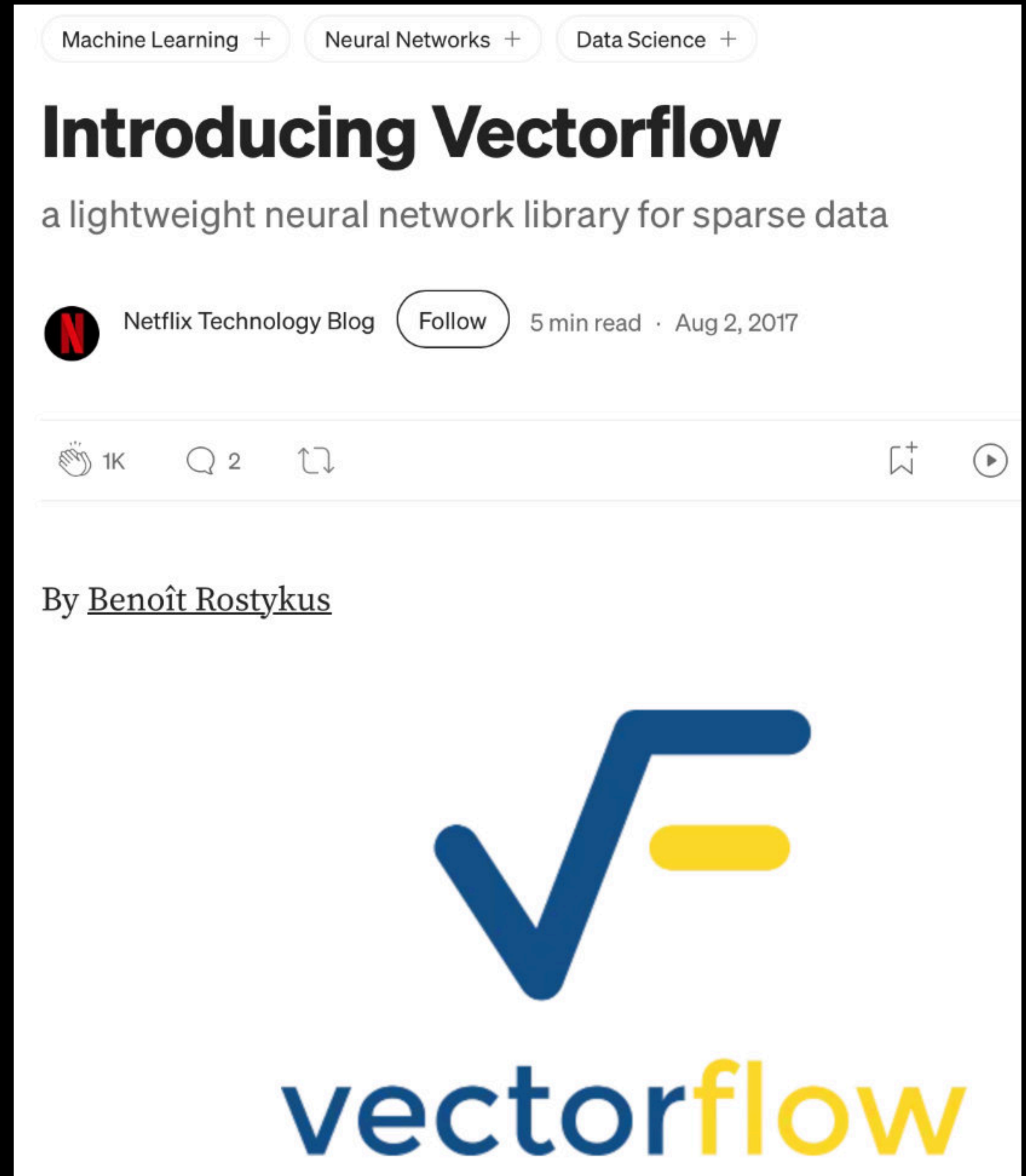
Discovery

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- Algorithms & Data Structures
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Discovery

- 3 semester educational program
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Discovery

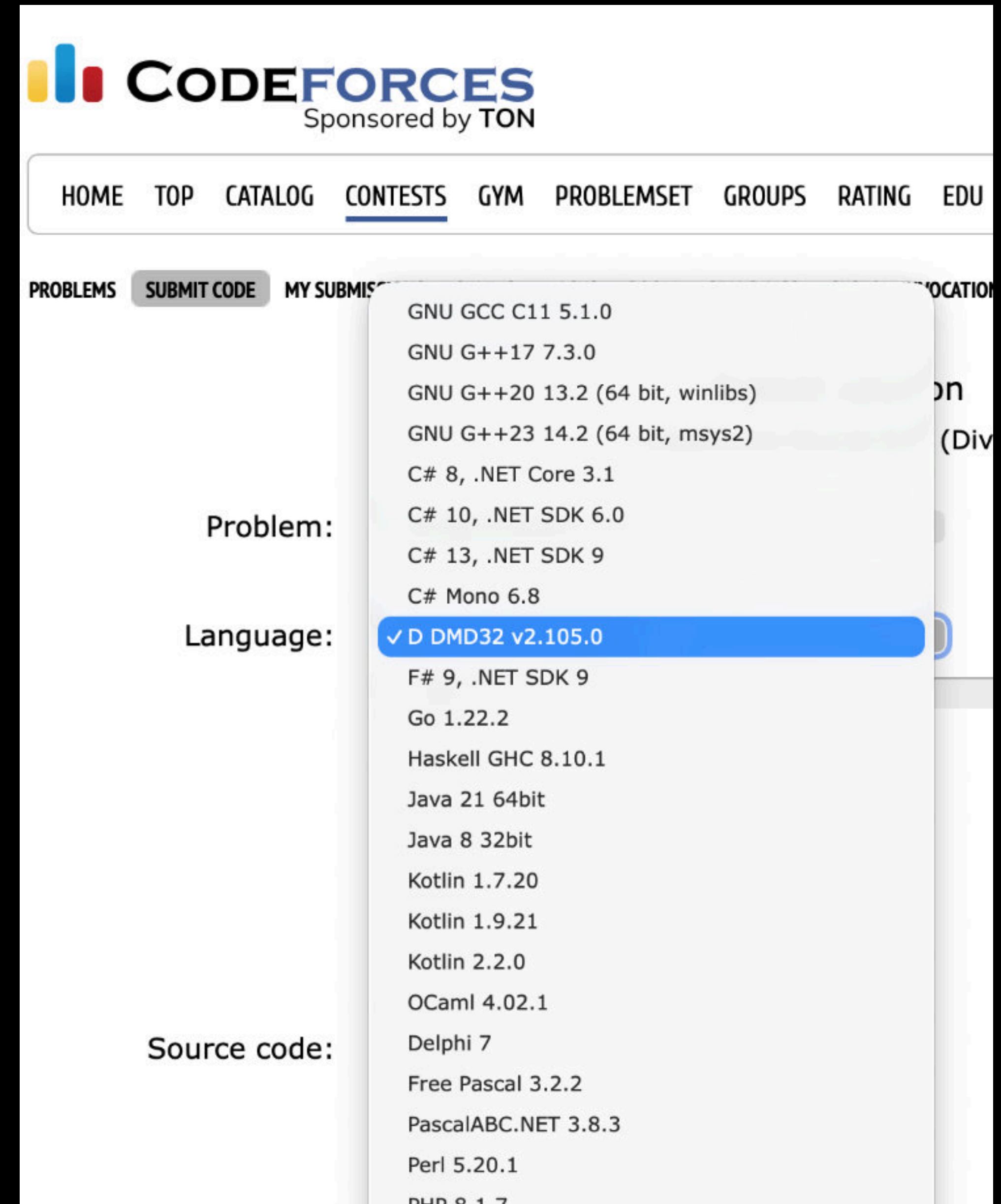
- 3 semester educational program
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Design considerations

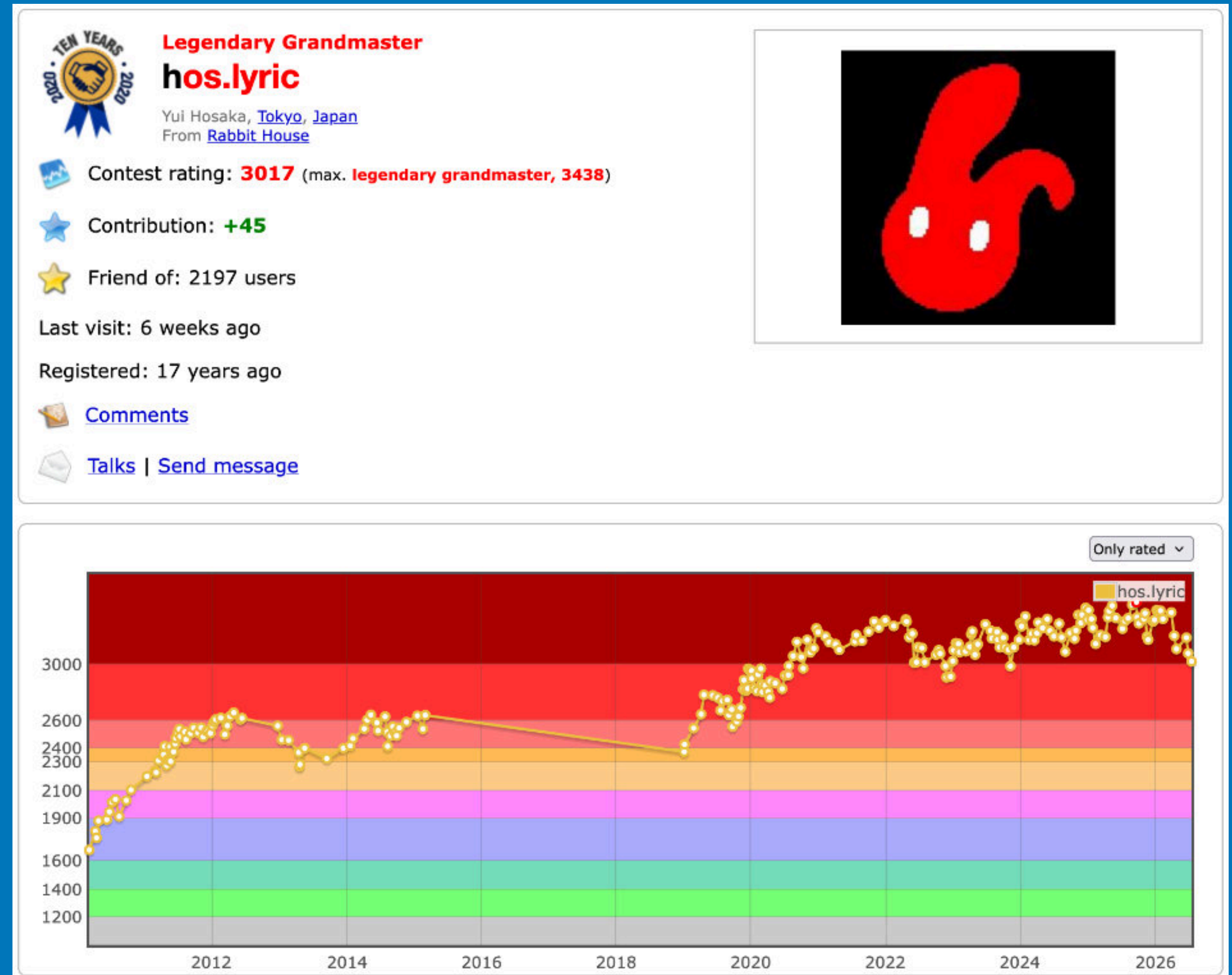
- **Agility.** We want our data scientists to easily run and iterate on their models in total autonomy. So we wrote Vectorflow in D, a modern systems language with a very gentle learning curve. Thanks to its fast compilers and functional programming features, it offers a Python-like experience for beginners but with typically multiple orders of magnitude of performance gain at run-time, while enabling seasoned developers to leverage an excellent templating engine, compile-time functionalities and lower-level features (C interface, inline assembler, manual memory management, auto-vectorization, ...). Vectorflow does not have any third-party dependencies, which eases its deployment. It offers a callback-based API to easily plug-in custom loss functions for training.

Discovery

- 3 semester educational program
- Algorithms & Data Structures
- 2 main languages
- Codeforces
- But one day..



Grandmasters Of Competitive programming



First steps



First steps

- Start with the Tour and join the community

So far, I've looked at the tour and the documentation from the official website, but the code looks more like C++, given the specifics of D. I'm not a programmer at all, but an analyst and data scientist. I usually work with Python/R. But here, for my studies, I had to solve problems on Codeforces. We learned C++ at university, but I haven't had a great feeling about it since then. I tried D and it seemed to work well. But I want to delve deeper.

First steps

- Start with the Tour and join the community
- Pay as you go
- Arrays / Containers / Ranges

Algorithms & ranges

[std.algorithm](#)

[std.range](#)

[std.range.primitives](#)

[std.range.interfaces](#)

Array manipulation

[std.array](#)

[std.algorithm](#)

Containers

[std.container.array](#)

[std.container.binaryheap](#)

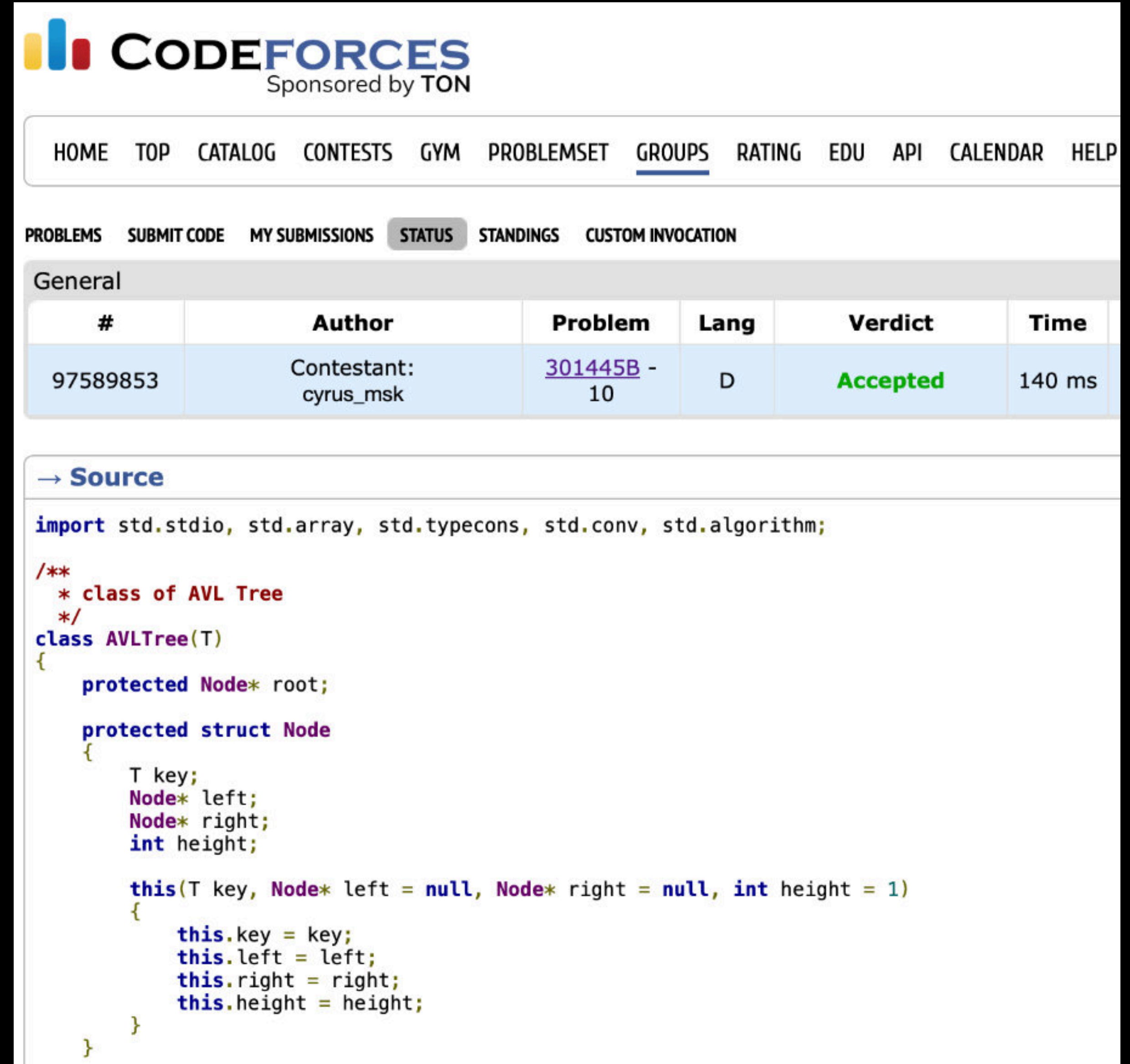
[std.container.dlist](#)

[std.container.rbtrees](#)

[std.container.slist](#)

First steps

- Start with the Tour and join the community
- Pay as you go
- Arrays / Containers / Ranges
- Read fast, write fast, run fast



The screenshot shows the Codeforces website interface. At the top is the Codeforces logo with the text "Sponsored by TON". Below the logo is a navigation bar with links: HOME, TOP, CATALOG, CONTESTS, GYM, PROBLEMSET, GROUPS, RATING, EDU, API, CALENDAR, and HELP. Below this is another navigation bar with links: PROBLEMS, SUBMIT CODE, MY SUBMISSIONS, **STATUS**, STANDINGS, and CUSTOM INVOCATION. The main content area is titled "General" and contains a table with submission details.

#	Author	Problem	Lang	Verdict	Time
97589853	Contestant: cyrus_msk	301445B - 10	D	Accepted	140 ms

Below the table is a link "→ Source" and a code editor showing C++ code for an AVL tree.

```
import std::stdio, std::array, std::typecons, std::conv, std::algorithm;

/**
 * class of AVL Tree
 */
class AVLTree(T)
{
    protected Node* root;

    protected struct Node
    {
        T key;
        Node* left;
        Node* right;
        int height;

        this(T key, Node* left = null, Node* right = null, int height = 1)
        {
            this.key = key;
            this.left = left;
            this.right = right;
            this.height = height;
        }
    }
}
```

First steps

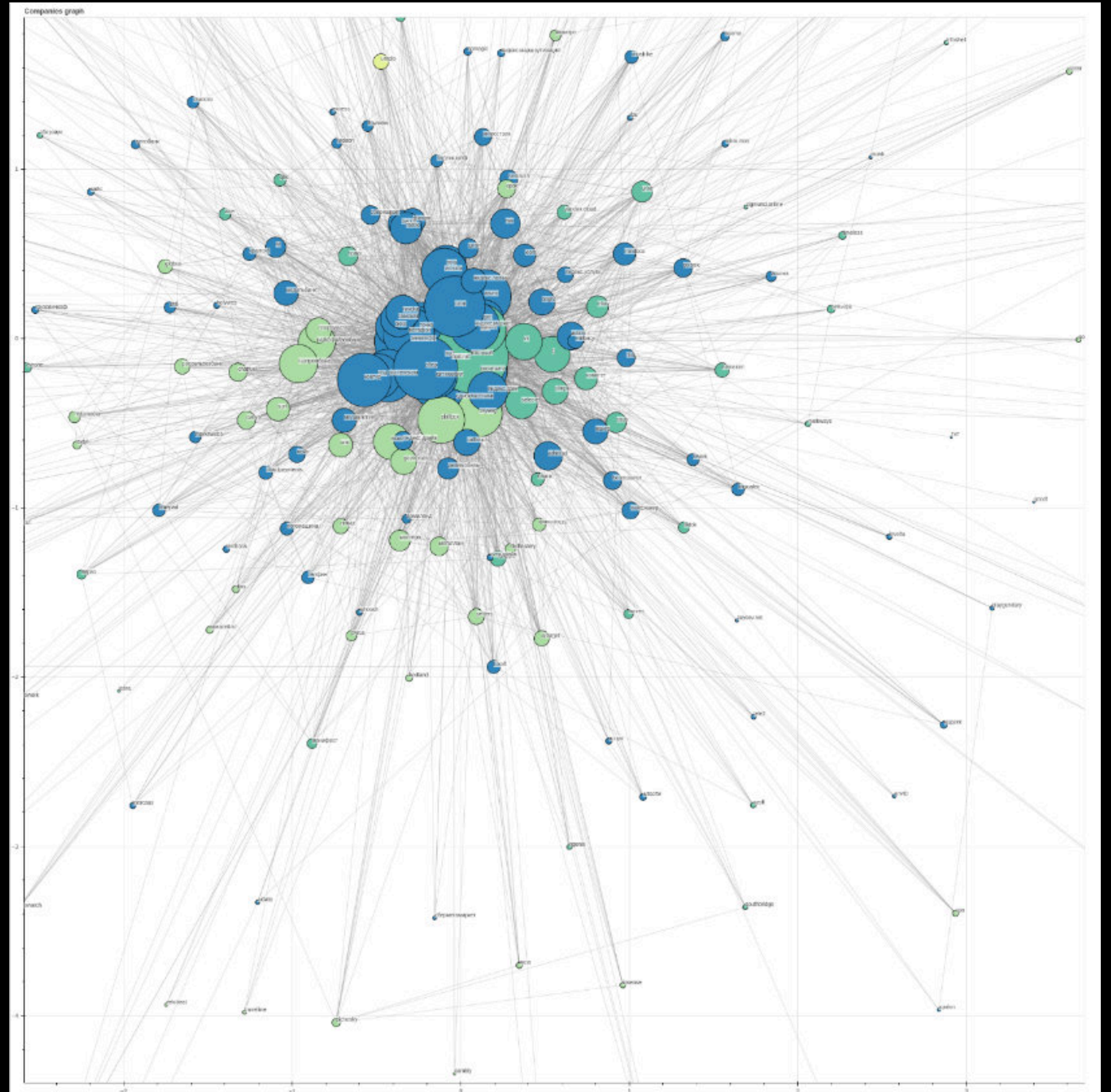
- Start with the Tour and join the community
- Pay as you go
- Arrays / Containers / Ranges
- Read fast, write fast, run fast
- Simple adoption, hard to fail

Programming Language	Avg. Errors to Solution	Tasks Solved
C	4,2	20
Python	3,3	59
Go	2,8	30
C++	2,7	61
Java	2,5	54
Scala	2,2	45
D	1,1	36
Others	0,9	8
All Languages	3,0	61

Programming Language	Submissions	Share
Python	36 075	62,5%
C++	20 371	35,3%
Java	465	0,8%
Go	223	0,4%
Scala	218	0,4%
C	207	0,4%
D	132	0,2%
Others	33	0,1%
Total	57 724	100%

First steps

- Start with the Tour and join the community
- Pay as you go
 - Arrays / Containers / Ranges
- Read fast, write fast, run fast
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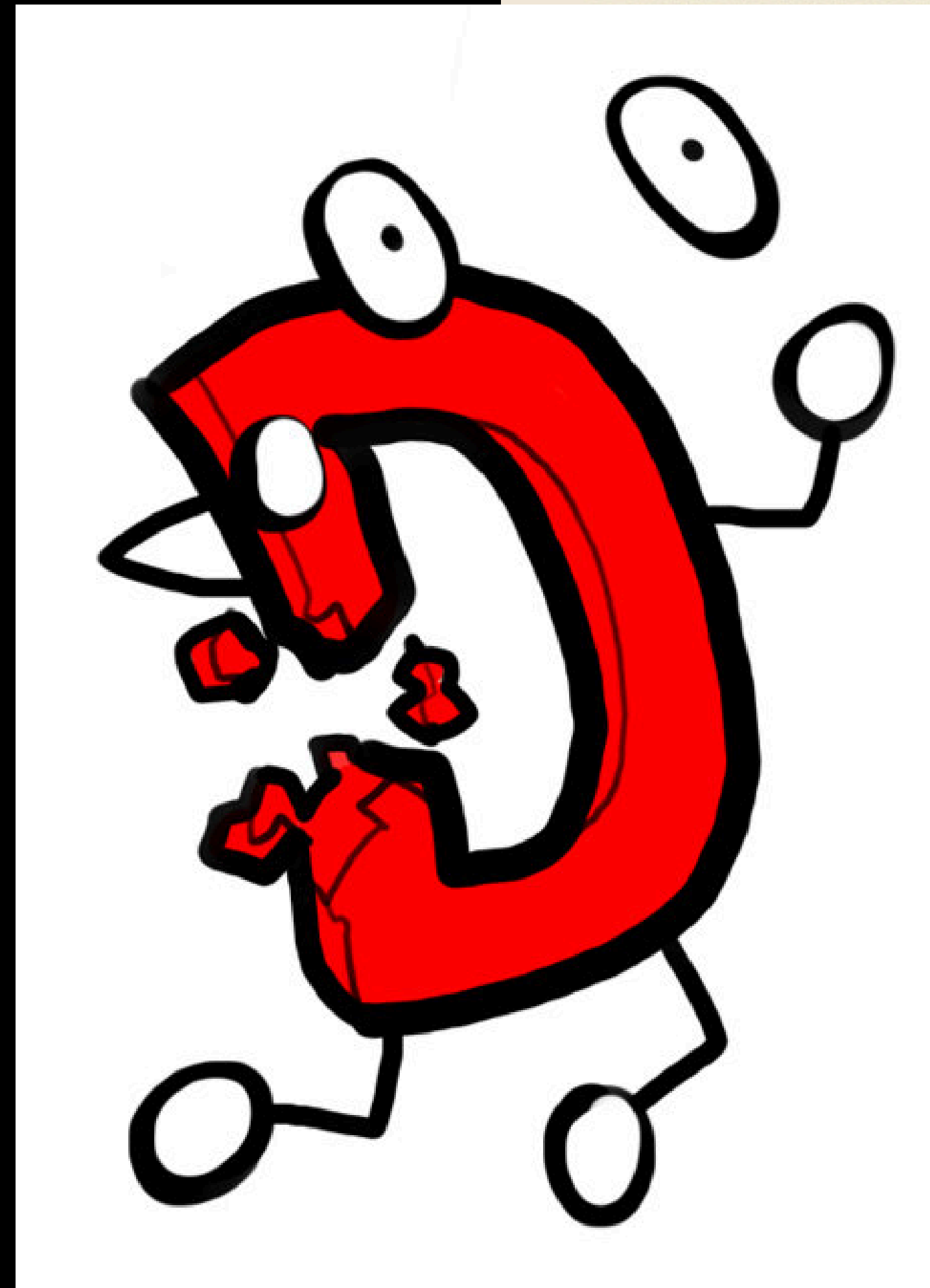


Growing confidence

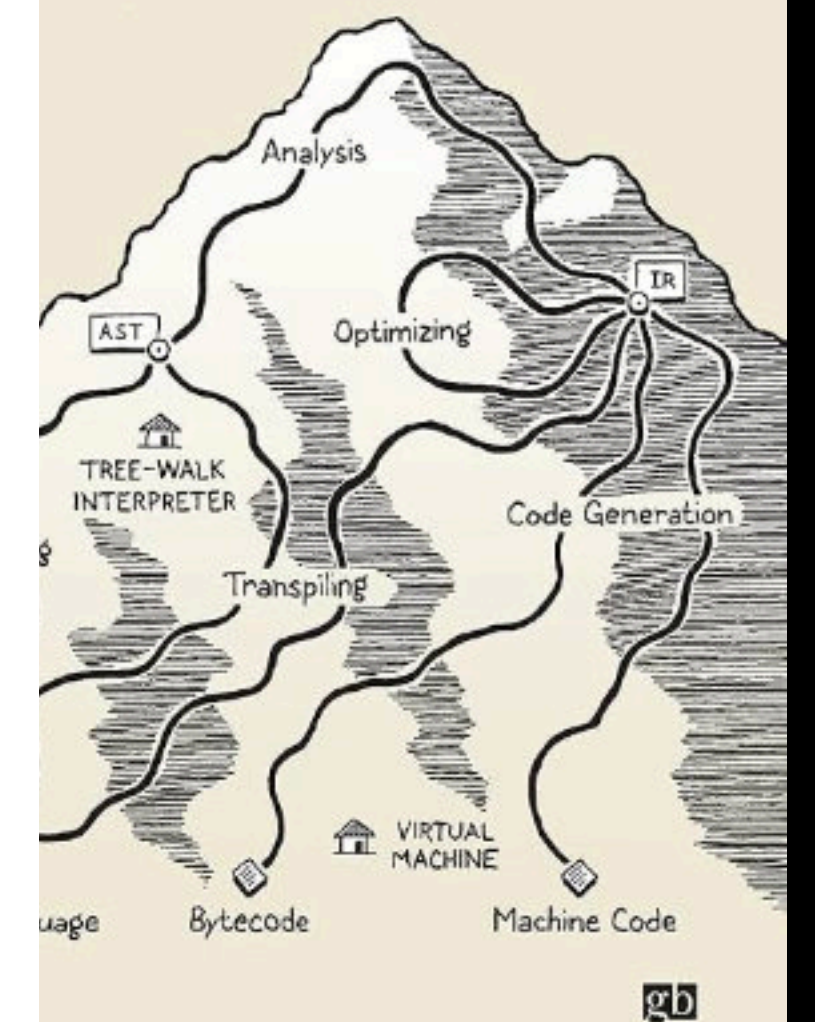


Growing confidence

- Small problems with existing solutions

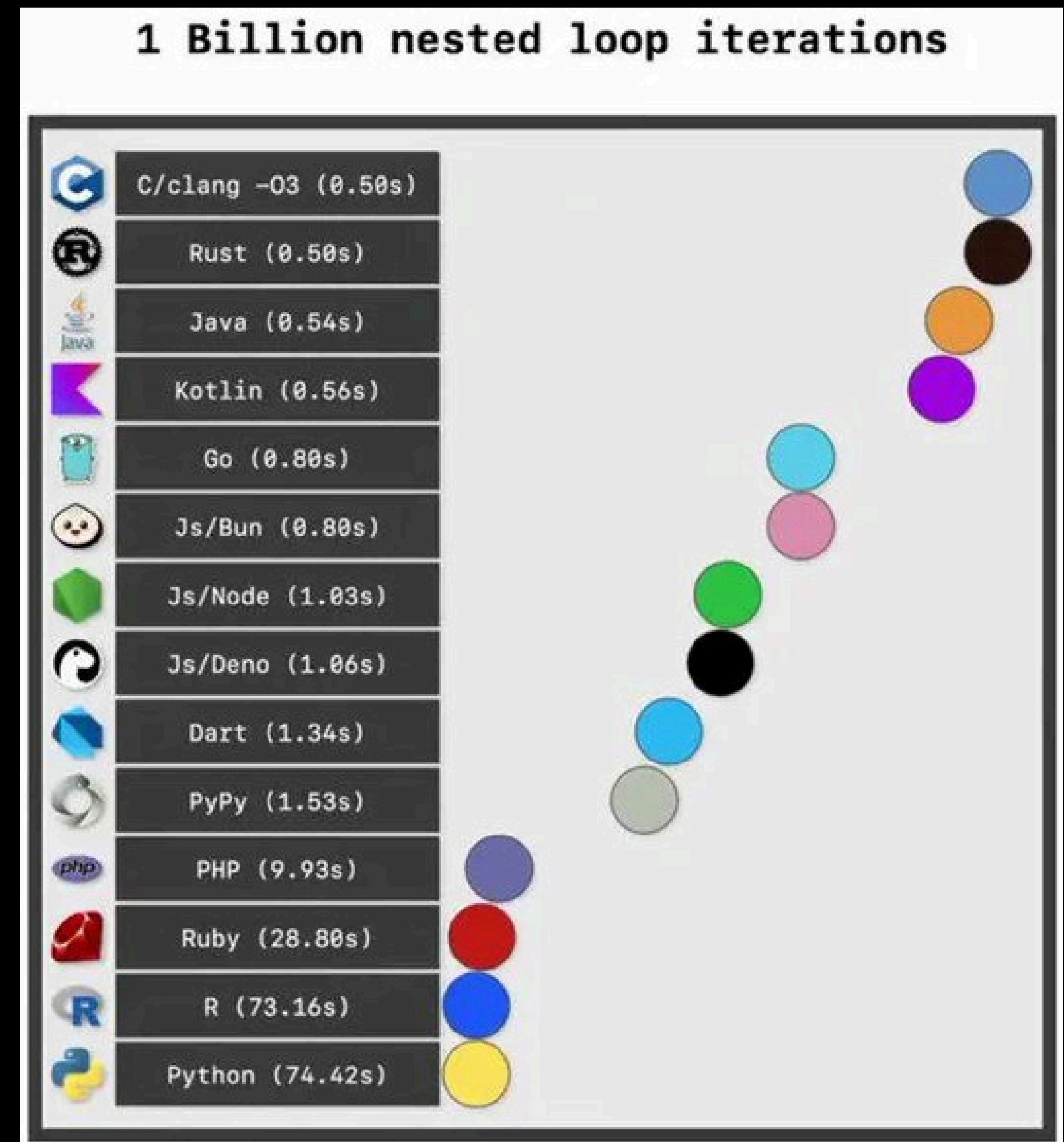


CRAFTING INTERPRETERS



Growing confidence

- Small problems with existing solutions
- Proper benchmarks are hard



Growing confidence

- Small problems with existing solutions
- Proper benchmarks are hard
- Different types and topics

Problems

[binarytrees](#)

[coro-prime-sieve](#)

[edigits](#)

[fannkuch-redux](#)

[fasta](#)

[helloworld](#)

[http-server](#)

[json-serde](#)

[knucleotide](#)

[lru](#)

[mandelbrot](#)

[merkletrees](#)

[nbody](#)

[nsieve](#)

[pidigits](#)

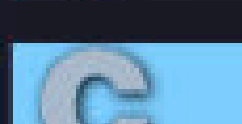
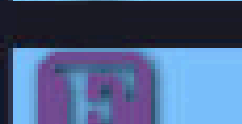
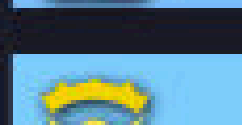
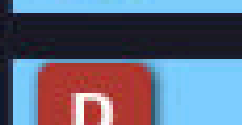
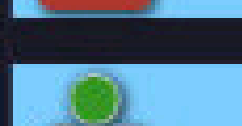
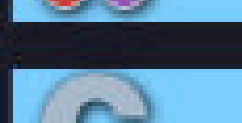
[regex-redux](#)

[secp256k1](#)

[spectral-norm](#)

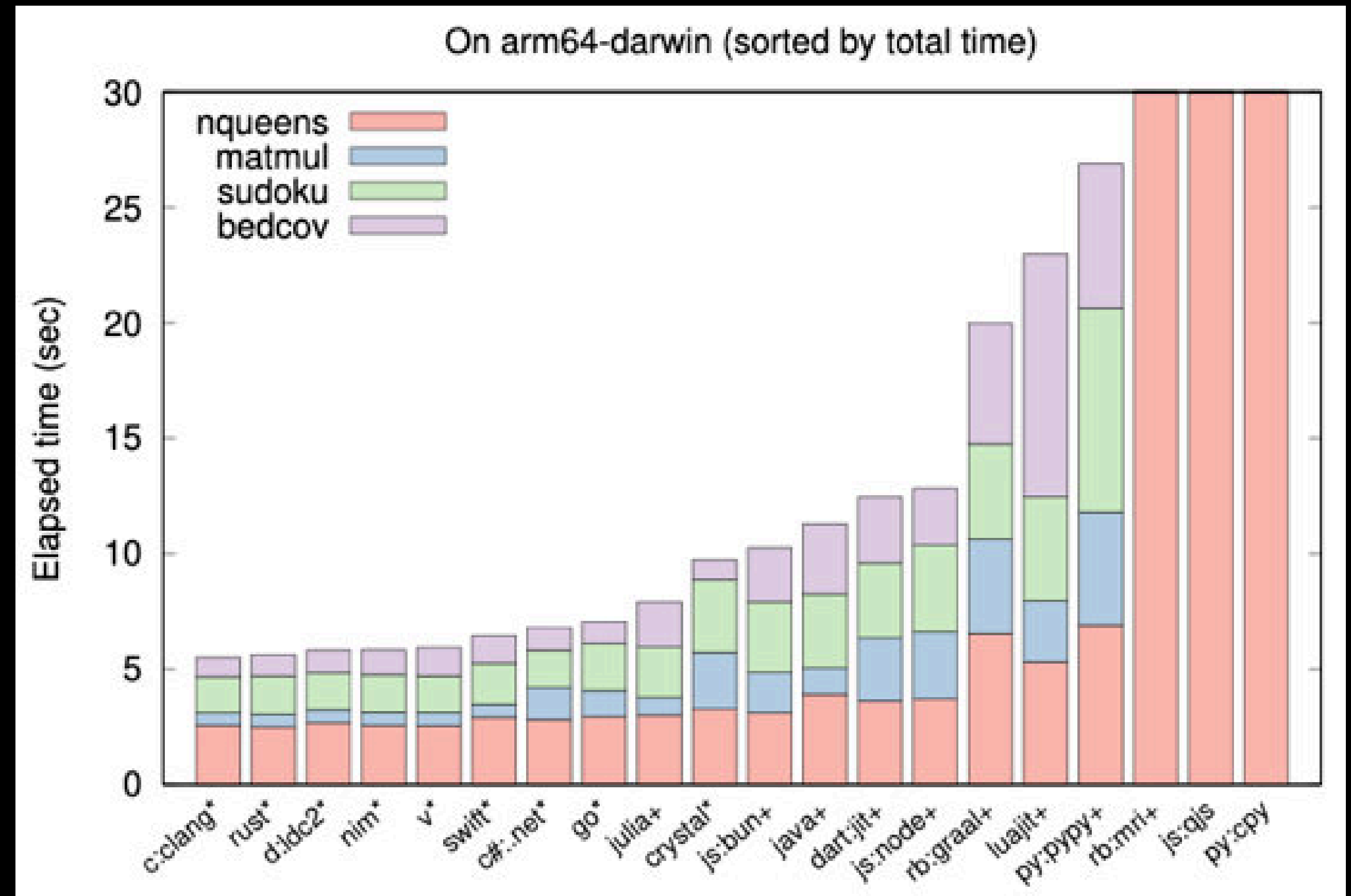
Growing confidence

- Small problems with existing solutions
- Proper benchmarks are hard
- Different types and topics:
 - Computationally intensive

C++ (clang++)	v21.1.2		223.0ms
C# (SIMD)	v10.0.101		225.1ms
C++ (g++)	v15.2.0		225.4ms
C++ (avx2)	v15.2.0		229.4ms
Zig	v0.15.2		232.6ms
D (LDC)	v1.41.0		232.9ms
Rust (nightly)	v1.94.0		233.8ms
C (clang)	v21.1.2		236.1ms
Fortran 90	v14.2.0		237.8ms
Nim	v2.2.0		238.8ms
D (GDC)	v15.2.0		242.3ms
Julia	v1.12.2		251.0ms
C (gcc)	v15.2.0		251.6ms

Growing confidence

- Small problems with existing solutions
- Proper benchmarks are hard
- Different types and topics:
 - Bit operations



Growing confidence

- Small problems with existing solutions
- Proper benchmarks are hard
- Different types and topics:
 - SIMD / parallel

fannkuch-redux

Input: 11

lang	code	time	stddev	peak-mem	compiler/runtime
d	1-im.d	374ms	6.5ms	3.0MB	ldc2 1.41.0
cpp	1-im.cpp	401ms	5.2ms	3.9MB	clang++ 14.0.0-1ubuntu1.1
cpp	1-im.cpp	402ms	3.8ms	2.6MB	g++ 11.4.0
rust	3-im.rs	413ms	0.7ms	2.1MB	rustc 1.88.0
rust	5-m.rs	558ms	2.0ms	2.0MB	rustc 1.88.0
zig	2-m.zig	639ms	3.6ms	1.4MB	zig 0.14.1
rust	2-im.rs	667ms	5.0ms	2.1MB	rustc 1.88.0
go	3-m.go	724ms	0.6ms	5.5MB	go 1.24.5
go	3-m.go	725ms	2.0ms	3.0MB	tinygo 0.38.0
java	1-m.java	901ms	5.9ms	96.3MB	graal/jvm 17.0.8

Growing confidence

- Small problems with existing solutions
- Proper benchmarks are hard
- Different types and topics:
 - Regex

Regex patterns

- Email: `[\w\.-]+@[\w\.-]+\.[\w\.-]+`
- URI: `[\w]+:\/[^\s?#]+[^\s?#]+(?:\[^\s#\]*|(?:[^\s]*))?(?:#[^\s]*)?`
- IPv4: `(?:((?:25[0-5]|2[0-4][0-9]|[01]?[0-9][0-9])\.){3}(?:25[0-5]|2[0-4][0-9]|[01]?[0-9][0-9])`

The above regex patterns aren't the best or the optimal. The focus is the benchmark, not the matching.

The patterns are applied to the whole file.

Measure

Measuring is done inside the programs to avoid include startup, reading and writing times on results.

Elapsed time include pattern compilation, find and count occurrences.

Growing confidence

- Small problems with existing solutions
- Proper benchmarks are hard
- Different types and topics:
 - Regex



Growing confidence

- Small problems with existing solutions
- Proper benchmarks are hard
- Different types and topics
- Community



Growing confidence

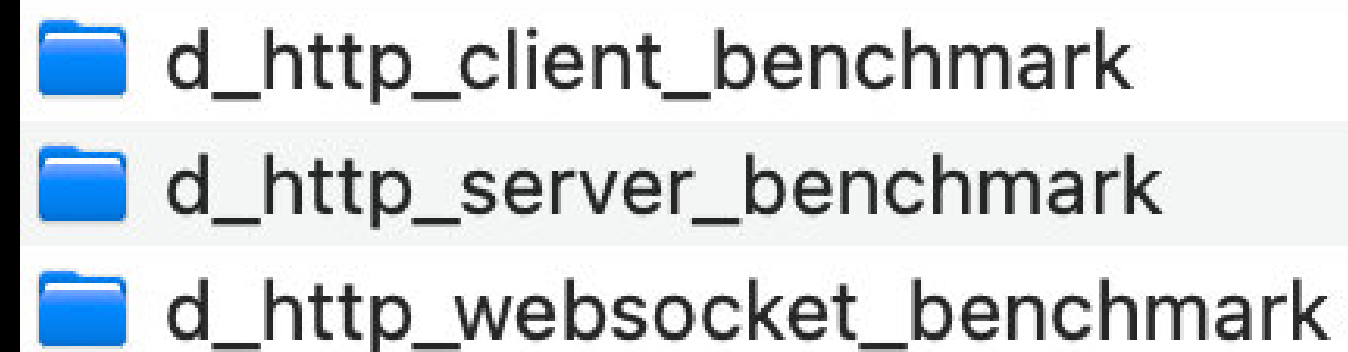
- Small problems with existing solutions
- Proper benchmarks are hard
- Different types and topics:
 - Web
 - Several server benchmarks

Language	Framework	Requests / Second (6...	↓ Requests / Second (2...	Requests / Second (5...
java (21)	activej (5.5)	178 628	152 251	153 227
java (21)	vertx (5.1)	166 709	142 038	138 415
d (2.112)	vibed_light (0.1)	154 069	132 598	133 321
csharp (14)	aspnet-minimal-api (10)	149 369	124 981	122 485
go (1.27)	gin (1.12)	131 788	111 955	109 969
d (2.112)	vibed (1.3)	135 614	109 172	104 652

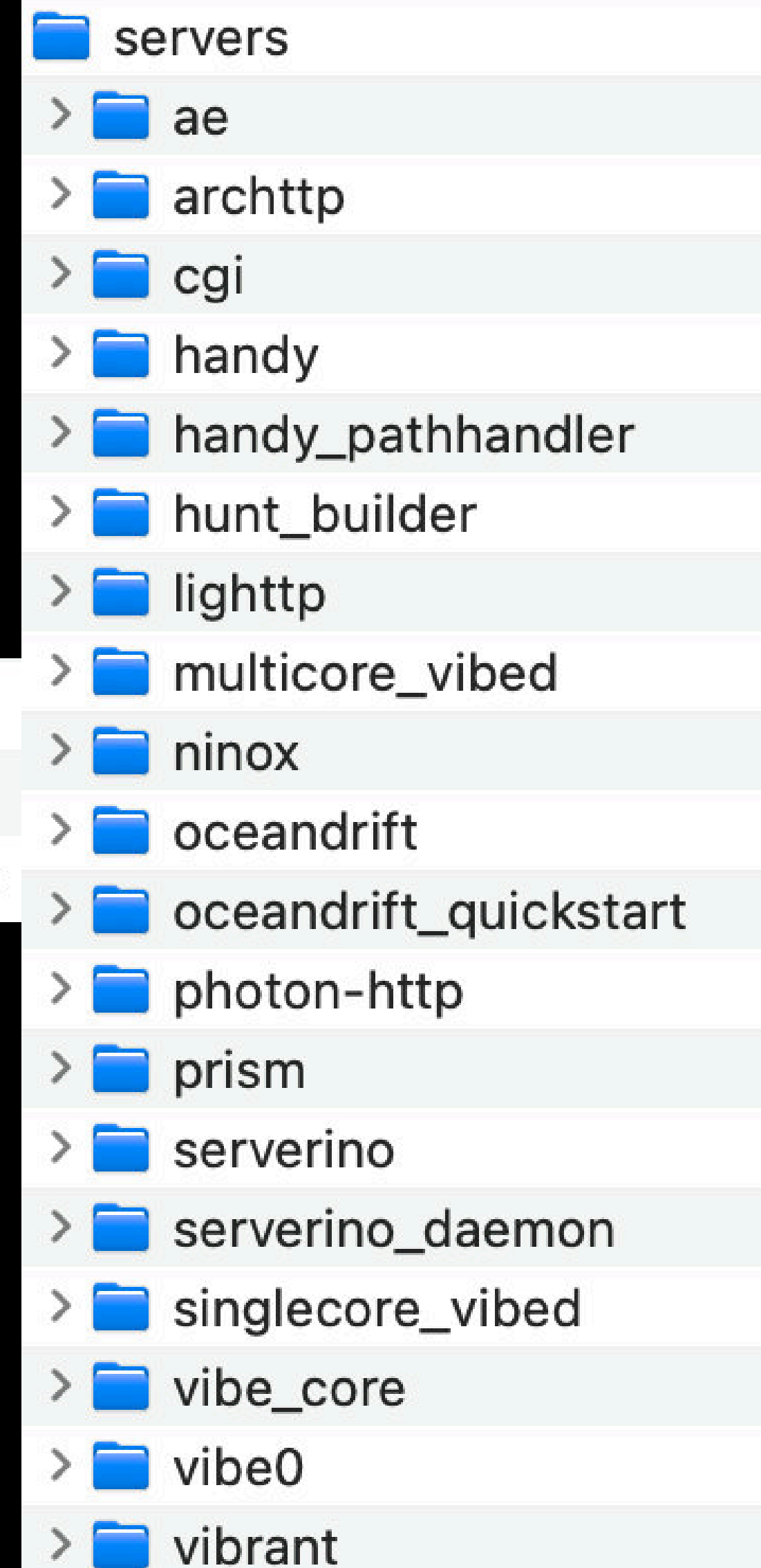
Framework	Requests per second (higher is better)	Latency (lower is better)	Stability (higher is better)	Memory (lower is better)
Java (Temurin) / Vert.x	23116	1.3 ms	66.75 %	484.4 MB
Rust / Actix	21965	1.4 ms	66.60 %	16.6 MB
C# (.NET) / ASP.NET Core	14707	1.2 ms	2.64 %	136.5 MB
D / Vibe.d	13037	2.4 ms	73.58 %	16.9 MB
Java (Temurin) / Spring Boot (WebFlux)	5813	0.8 ms	0.86 %	634.2 MB
Go / FastHTTP	5567	0.7 ms	0.82 %	13.4 MB
Kotlin (Temurin) / Ktor (Netty)	2613	1.3 ms	1.37 %	557.3 MB
Java (Temurin) / Spring Boot (MVC)	1105	1.7 ms	1.67 %	596.8 MB

Growing confidence

- Small problems with existing solutions
- Proper benchmarks are hard
- Different types and topics:
 - Web
 - My own comparisons



d_http_client_benchmark
d_http_server_benchmark
d_http_websocket_benchmark



servers

- > ae
- > archttp
- > cgi
- > handy
- > handy_pathhandler
- > hunt_builder
- > lighttp
- > multicore_vibed
- > ninox
- > oceandrift
- > oceandrift_quickstart
- > photon-http
- > prism
- > serverino
- > serverino_daemon
- > singlecore_vibed
- > vibe_core
- > vibe0
- > vibrant

Growing confidence

- Small problems with existing solutions
- Proper benchmarks are hard
- Different types and topics:
 - Web
 - GraphQL

Results

Name	Language	Server	Latency avg	Requests
static-rust	Rust	Actix Web	1.79ms	110kps
graphql-crystal	Crystal	Kemal	2.93ms	68kps
gqlgen	Go	net/http	3.67ms	54kps
async-graphql	Rust	Actix Web	4.44ms	45kps
Juniper	Rust	Actix Web	5.16ms	39kps
Hot Chocolate	C#	ASP.NET	8.14ms	25kps
Mercurius	Node.js	Fastify	8.55ms	23kps
graphql-go	Go	net/http	9.96ms	20kps
GraphQLD	D	Vibe.d	11.04ms	18kps
agoo	Ruby/C	agoo	12.53ms	16kps
graphql-yoga	Node.js	http	13.66ms	15kps
graphql-jit	Node.js	http	19.11ms	10kps
Absinthe	Elixir	Phoenix	23.16ms	8.6kps
Hono	Bun	HonoJS	27.55ms	7.3kps

Web Frameworks In D

- Arsd CGI @Adam
- Handy-web @Andrew
- Serverino @Andrea

Web Frameworks

Networking libraries.

- [dlang-requests](#) - HTTP client library inspired by python-requests
- [Handy-Httpd](#) - A simple, lightweight, and well-documented HTTP server that lets you bootstrap ideas and have something up and running in minutes.
- [serverino](#) - Small and ready-to-go http server, in D
- [libasync](#) - Cross-platform event loop library of asynchronous objects
- [libhttp2](#) - HTTP/2 library in D, translated from nghttp2

Full stack web frameworks.

- [vibe.d](#) - Asynchronous I/O Web Framework that doesn't get in your way, written in D.
- [arsd](#) - Adam D. Ruppe's web framework. (See `arsd/cgi.d` for the CGI/FastCGI/SCGI/webserver component.)
- [cmsed](#) - A component library for Vibe that functions as a CMS.

One last benchmark In D

- Started with an initial implementation



Serg Gini @cyrus_msk · Oct 14, 2023

D is slow they said.. it is using GC they said

#dlang

github.com/jinyus/related...

VM Specs (Azure F4s v2 - 4vCPU-8GB-Ubuntu 22.04)

Language	Time (5k posts)	20k posts	60k posts	Total
D	23.84 ms	302.34 ms	2.63 s	2.95 s
Go	24.41 ms	370.44 ms	3.28 s	3.67 s
Rust	23.85 ms	387.30 ms	3.45 s	3.86 s
Zig	28.08 ms	427.33 ms	3.82 s	4.28 s
F# (AOT)	32.31 ms	486.67 ms	4.14 s	4.66 s
Nim	43.03 ms	398.08 ms	4.96 s	5.40 s
Julia	32.23 ms	574.00 ms	5.05 s	5.65 s
Java (GraalVM)	33.08 ms	495.33 ms	5.29 s	5.82 s
C# (AOT)	42.77 ms	616.74 ms	5.42 s	6.08 s
F#	44.85 ms	623.67 ms	5.52 s	6.19 s
Odin	41.30 ms	575.45 ms	6.07 s	6.69 s

3

12

43

3K

One last benchmark

In D

- Started with an initial implementation
- Bruce Carneal in D house

VM Specs (AWS c7a.xlarge-4vCPU-8GB-RAM-Ubuntu 24.04)				
Language	Time (5k posts)	20k posts	60k posts	Total
D (v2)	12.06 ms	122.21 ms	961.08 ms	1.10 s
Rust	9.84 ms	124.59 ms	1.10 s	1.23 s
c3	10.50 ms	141.00 ms	1.13 s	1.28 s
C++	12.20 ms	164.33 ms	1.39 s	1.57 s
Zig	15.00 ms	199.33 ms	1.71 s	1.93 s
Odin	14.87 ms	208.01 ms	1.78 s	2.00 s
Go	16.64 ms	240.66 ms	2.12 s	2.37 s
Java (JIT)	20.80 ms	246.00 ms	2.12 s	2.39 s
Neat	19.01 ms	254.22 ms	2.15 s	2.42 s
Haskell	21.10 ms	277.44 ms	2.24 s	2.54 s
Nim	18.23 ms	259.83 ms	2.29 s	2.57 s
F# (AOT)	18.68 ms	266.89 ms	2.33 s	2.61 s
C# (JIT)	21.20 ms	296.19 ms	2.48 s	2.80 s

Ports & bindings



Ports & Bindings

- One more step before you go wild



Ports & Bindings

- One more step before you go wild
- Ports
- Dirsplitter CLI
- Knowledge
- CLI libraries

```
dirsplitter -h
```

USAGE:

USAGE

```
$ dirsplitter [-h] [--version] split|reverse
```

FLAGS

-h, --help	prints help
--version	prints version

SUBCOMMANDS

split	Split a directory into parts of a given size
reverse	Reverse a split directory

SPLIT USAGE:

```
dirsplitter split: Split a directory into parts of a given size
```

USAGE

```
$ dirsplitter split [-h] [-m MAX] [-d TARGET] [-p PREFIX]
```

FLAGS

-h, --help	prints help
------------	-------------

OPTIONS

-m, --max MAX	Max part size in GB [default: 5.0]
-d, --dir TARGET	Target directory
-p, --prefix PREFIX	Prefix for output files of the tar command.

Ports & Bindings

- One more step before you go wild
- Ports
- tzf CLI
- Knowledge
- timezones databases

tzf-rs: a fast timezone finder for Rust.

 Rust passing docs passing crates.io v1.3.7  license scan passing



Dates and UTF

In D

- Advent of Code on Internationalization and Dates
- <https://i18n-puzzles.com/>

Internationalization Puzzles

[Puzzles](#) | [About](#) | [Leaderboard](#) | [Stats](#) | [GitHub Login](#) | [Google Login](#)

This is a series of **programming puzzles**, designed to teach you about internationalization... [Read on...](#)

Join the discussion on [Discord](#) or [Reddit](#). Sign up for [updates by email](#)

Puzzles

1	--	Puzzle Available	--	-
2	--	Puzzle Available	--	-
3	--	Puzzle Available	--	-
4	--	Puzzle Available	--	-
5	--	Puzzle Available	--	-
6	--	Puzzle Available	--	-
7	--	Puzzle Available	--	-
8	--	Puzzle Available	--	-
9	--	Puzzle Available	--	-
10	--	Puzzle Available	--	-
11	--	Puzzle Available	--	-
12	--	Puzzle Available	--	-
13	--	Puzzle Available	--	-
14	--	Puzzle Available	--	-
15	--	Puzzle Available	--	-
16	--	Puzzle Available	--	-
17	--	Puzzle Available	--	-
18	--	Puzzle Available	--	-
19	--	Puzzle Available	--	-
20	--	Puzzle Available	--	-

Ports & Bindings

- One more step before you go wild
- Ports
- ddiff CLI
- Knowledge
- work with picture format



The fastest* (one-thread) pixel-by-pixel image difference tool in the world.

MADE WITH

REASON

GLUTEN

FREE

POWERED BY

OVERTIME

Why Odif?

ODiff is a blazing fast native image comparison tool. Check [benchmarks](#) for the results, but it compares the visual difference between 2 images in **milliseconds**. ODiff is designed specifically to handle significantly similar images like screenshots, photos, AI-generated images and many more. ODiff is designed to be portable, fast, and memory efficient.

Originally written in OCaml, currently in Zig with SIMD optimizations for SSE2, AVX2, AVX512, and NEON.

Ports & Bindings

- One more step before you go wild

- Ports

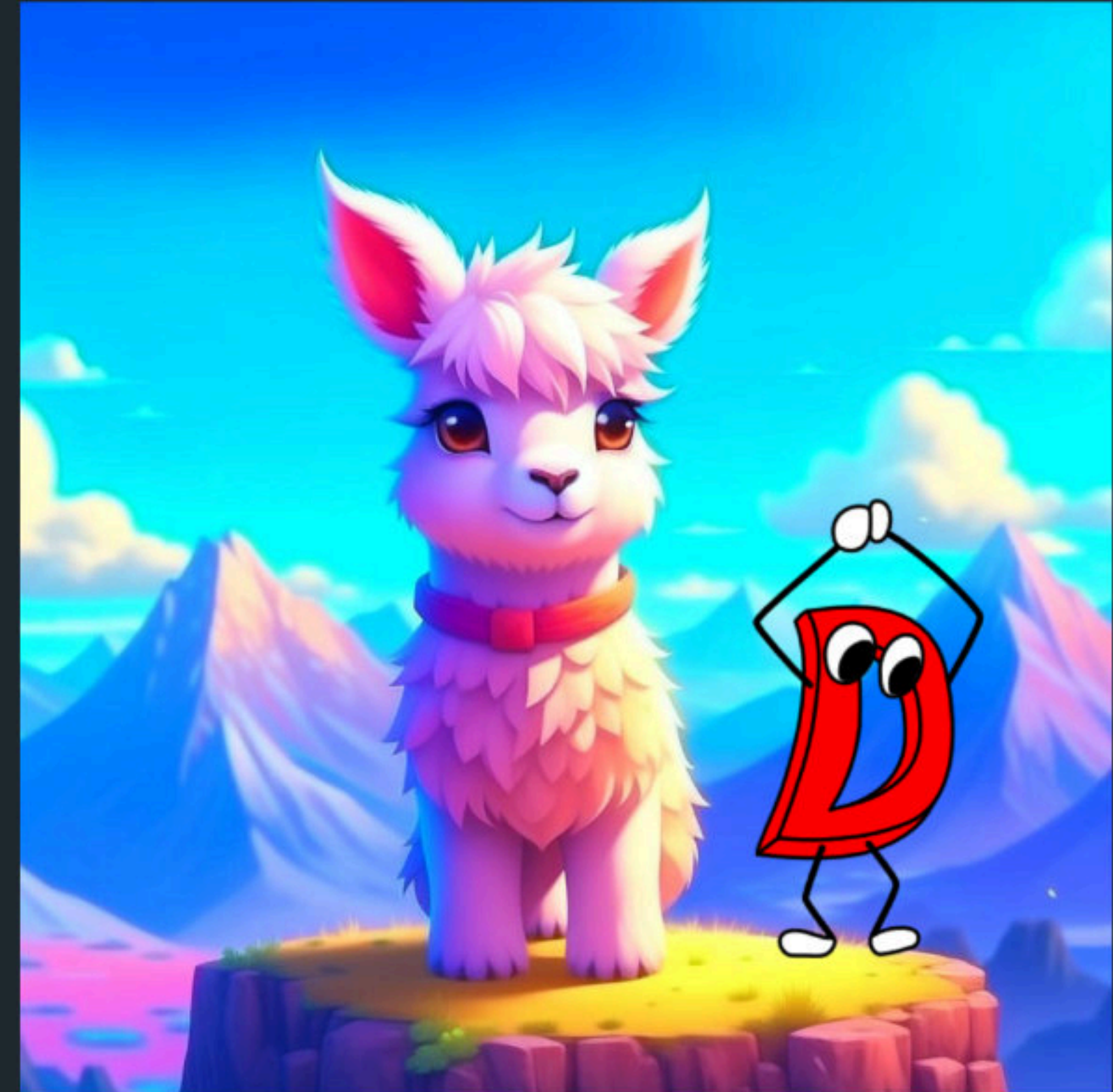
- Karpathy's work

- microgpt

- llm.c

- llama2.c

llama2.d



This is the D version of [llama2.c](#) by Andrej Karpathy. It runs inference for the [llama2](#) model architecture recently published by Meta.

Ports & Bindings

- One more step before you go wild
- Ports
- Bindings

Add ImportC compiler to dmd

May 09, 2021

Walter Bright



 [Permalink](#)
 [Reply](#)
 [Subscribe](#)

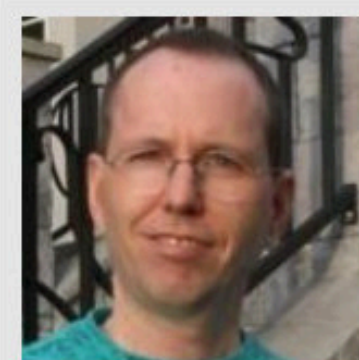
<https://github.com/dlang/dmd/pull/12507>

If you could add a C compiler to dmd with 3000 lines of code, so C code could be imported directly? I would!

Re: Add ImportC compiler to dmd

May 09, 2021

Walter Bright



Posted in reply to
[Walter Bright](#)

 [Permalink](#)
 [Reply](#)

My original plan was to use DMC for this. But looking at the code to it dissuaded me. It would have been more work than just starting over again, particularly because only the lexer/parser were needed.

Many times people over the years have asked me to port DMC to Linux. Ironically, this actually does that!

Ports & Bindings

- One more step before you go wild
- Ports
- Bindings
 - `igraph-d`



igraph – The network analysis package

igraph is a collection of network analysis tools with the emphasis on efficiency, portability and ease of use. igraph is open source and free. igraph can be programmed in R, Python, Mathematica and C/C++.

igraph R package

python-igraph

IGraph/M

igraph C library

Ports & Bindings

- One more step before you go wild
- Ports
- Bindings
 - plutovg-d

```
{
  "libs": [
    "plutovg"
  ],
  "lflags-posix": [
    "-L$PACKAGE_DIR/pluto_lib/object"
  ],
  "sourcePaths": ["source"],
  "cSourcePaths": ["pluto_lib"],
  "dflags-ldc": ["-mcpu=native"],
  "lflags-osx": ["-rpath", "@executable_path/"],
  "authors": [
    "Sergei Giniatulin"
  ],
  "targetType": "staticLibrary",
  "copyright": "Copyright © 2025, Sergei Giniatulin",
  "description": "Minimal example for Plutovg in D",
  "license": "MIT",
  "name": "plutovg_d"
}
```

Sorting algorithms

In D

- Research project on sort algorithms for Rust

sort-research-rs

- Is a sort implementation correct?
- Is a sort implementation fast?

This repository contains:

- An exhaustive test suite, including [properties](#) not commonly checked or upheld
- An extensive benchmark suite, abstracting over types, patterns and sizes
- A fuzzing harness
- A novel sort implementation [ipnsort](#) (Instruction-Parallel-Network-Sort)
- Vended sort implementations (Rust, C++, C), e.g. `cpp_pdqsort`, `rust_std_stable`
- Various experiments and demonstrations
- Results of the research as [papers](#)

Most tests and benchmarks can be applied to non Rust implementations. This works by implementing the 5 benchmark types as `#[repr(C)]` and having a defined C API that can be implemented, wrapping the C/C++ sorts.

Most functionality is by default disabled via cargo features, see the Cargo.toml. Some functionality can be enabled or switched by setting environment variables. See for example `benches/bench.rs`.

Sorting algorithms In D

- Research project on sort algorithms for Rust
- Successfully ported
 - Zed extension by @gdamore
 - LLDB help by @webfreak

```
sort_test
├── .zed
├── dev_test
├── source/drift_sort
│   ├── drift.d
│   ├── merge.d
│   ├── package.d
│   ├── pivot.d
│   └── quicksort.d
├── smallsort.d
├── .gitignore
├── dub.json
├── sort_test
└── test
```

```
source/drift_sort/quicksort.d › void stable_quick_sort(T, alias less = "a < b")(
    T[] v, T[] scratch, uint limit, Nullable!(T*) left,
)
{
    pragma(inline, false)
    void stable_quick_sort(T, alias less = "a < b")(
        T[] v, T[] scratch, uint limit, Nullable!(T*) left,
    )
    {
        alias is_less = binaryFun!less;
        while (true)
        {
            auto len = v.length;
            if (len <= 1)
            {
                return;
            }
            pivot_pos = 0;
            pivot_val = v[0];
            left_ancestor_pivot = { _isNull: true };
            limit = 12;
            scratch = { length: 512, ptr: 0x00000000 };
            v = { length: 100, ptr: 0x00000001 };
        }
    }
}
```

1: tid=294861 • Dub Debug (macOS)

Frames	Breakpoints	Console	Variables	Terminal
quicksort.d: smallsort.d: C++: on cat C++: on thr			Local v = {length:100, ptr:0x00000001} scratch = {length:512, ptr:0x00000000} limit = 12 left_ancestor_pivot = { _isNull: true } len = 6171913210 pivot_pos = 0 pivot_val = {length:0, ptr:0x00000000}	Result std: 1158 []

Personal projects



Imagination is your limit

- Advent of Code

```
AoC Day 6 Part 2 in D language

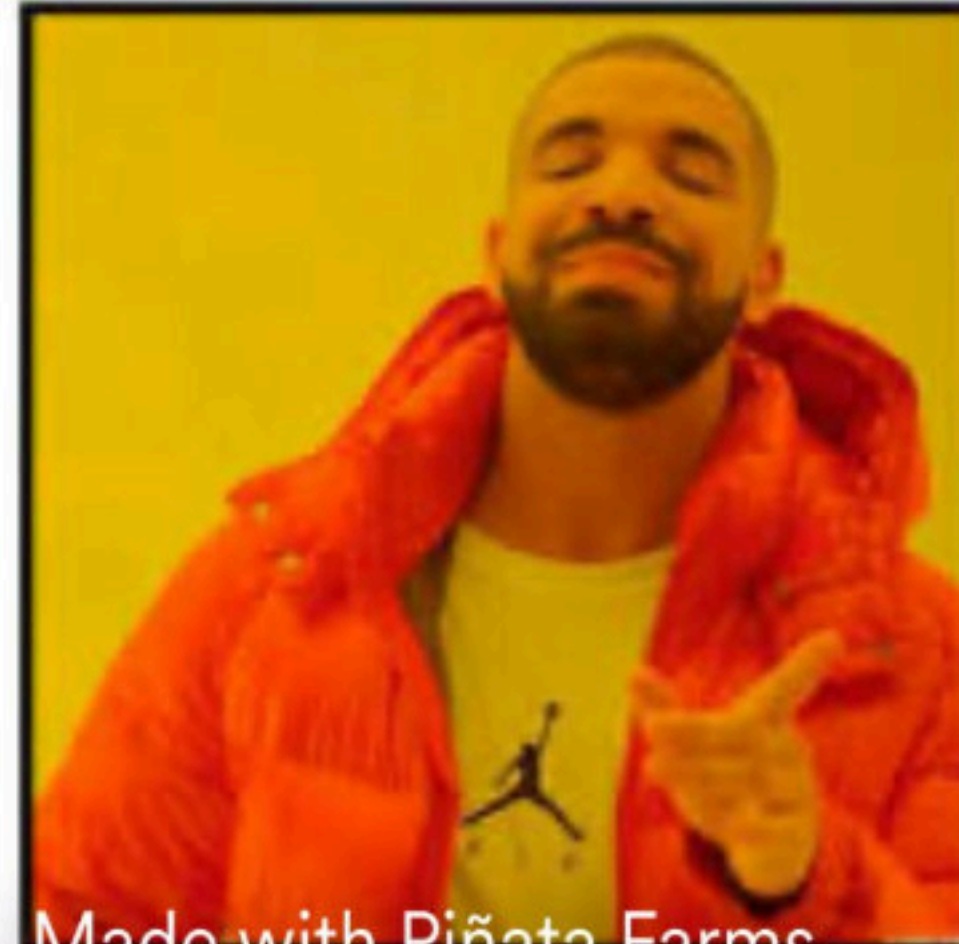
1  auto data = readText("input.txt").strip;
2  auto first_op = min(data.countUntil("*"), data.countUntil("+"));
3  auto operands_line = data[first_op .. $];
4  data = data[0 .. first_op - 1];
5
6  long result;
7  long[][long] table;
8
9  long i = 0;
10 foreach(element; data.split("\n").array.transposed)
11 {
12     auto el = element.to!string.strip;
13     if (el == "")
14         i++;
15     else
16         table[i] ~= el.to!long;
17 }
18 foreach(k, operand; operands_line.strip.split)
19 {
20     if (operand == "+")
21         result += (table[k]).fold!"a + b";
22     else
23         result += (table[k]).fold!"a * b";
24 }
25 writeln(result);
```

Imagination is your limit

- Advent of Code



**Implement union–find
data structure to find all
groups**

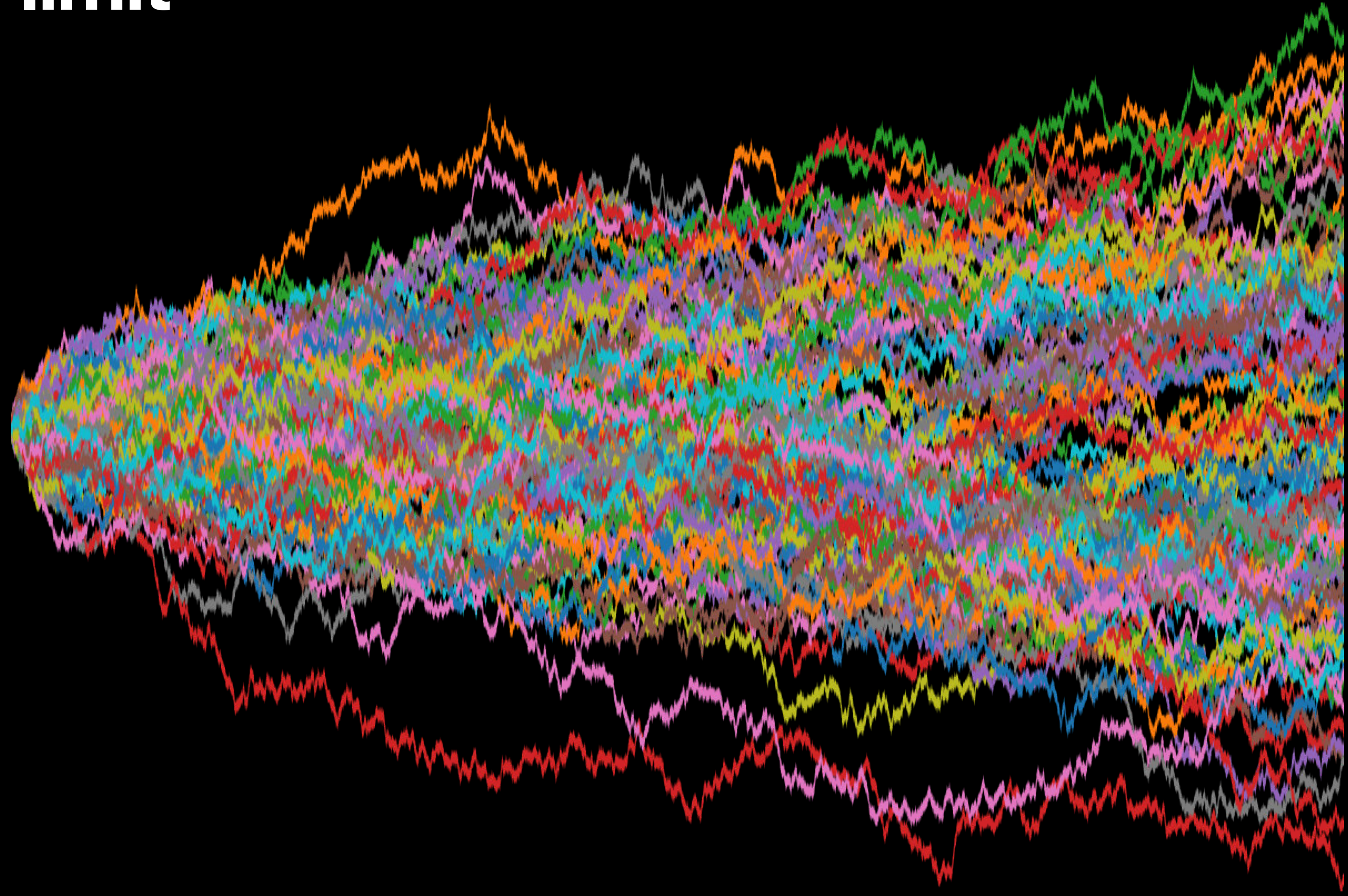


**Use graph library
to identify all
graph's components**

Made with Piñata Farms

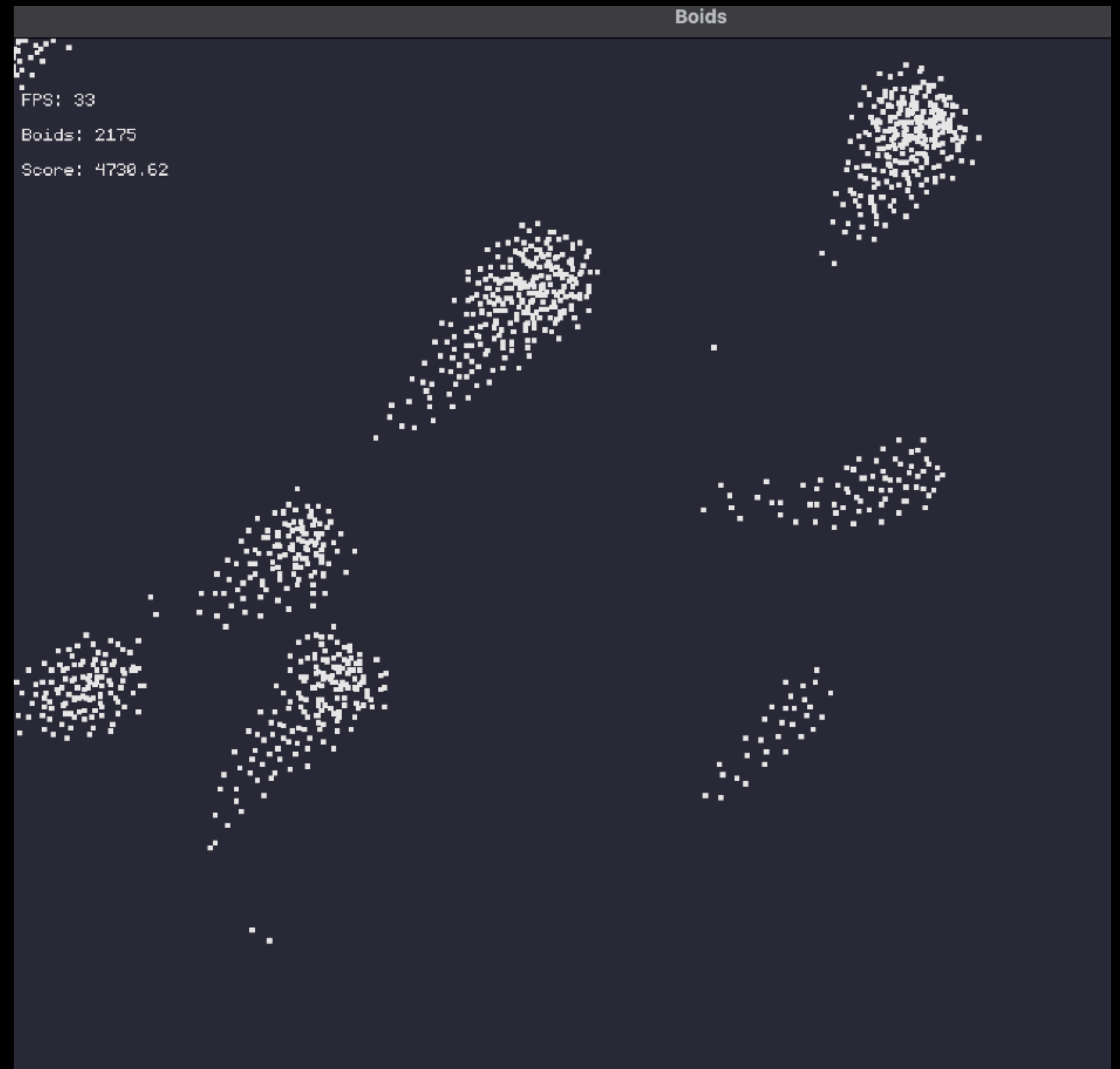
Imagination is your limit

- Advent of Code
- Games / graphics



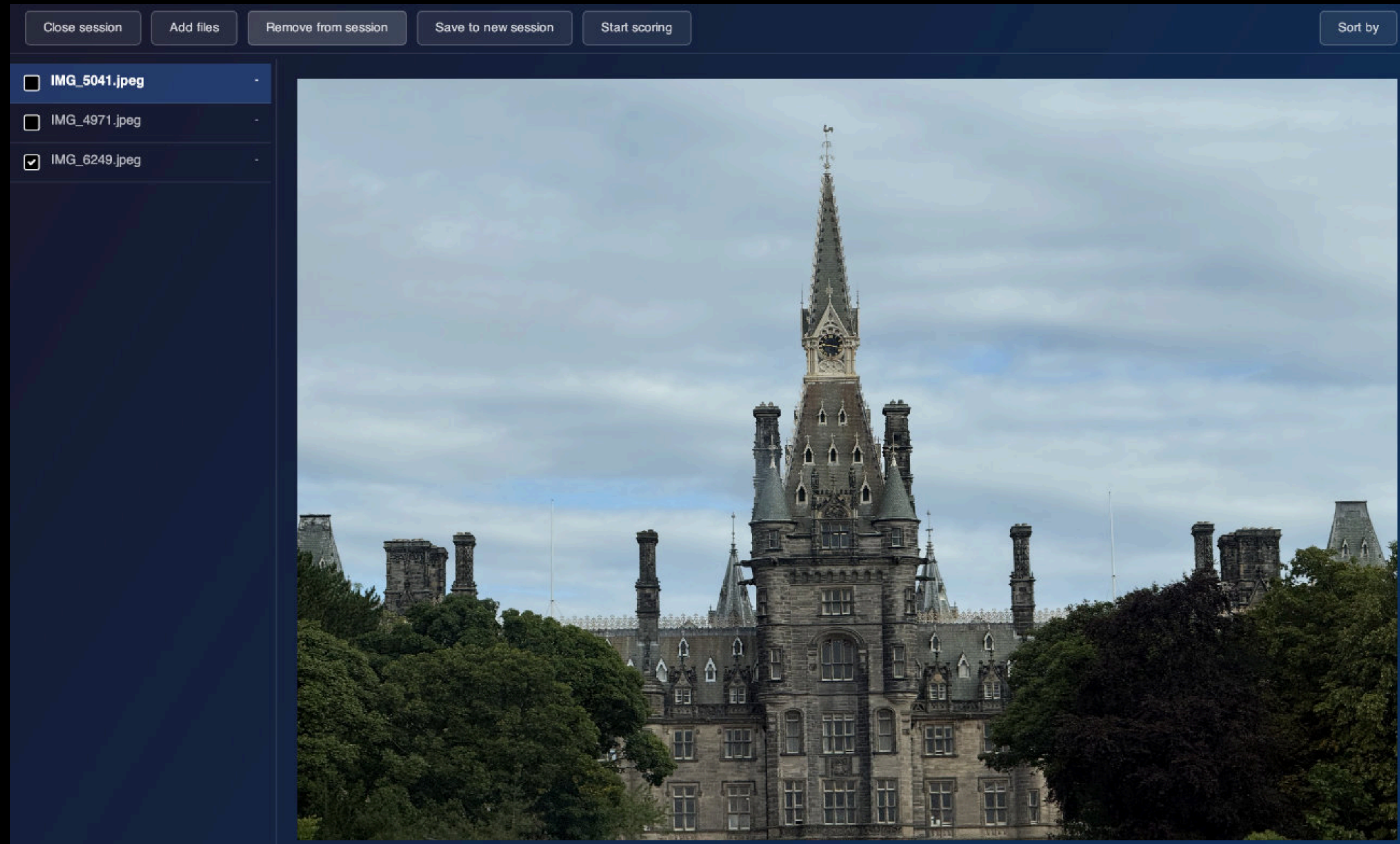
Imagination is your limit

- Advent of Code
- Games / graphics



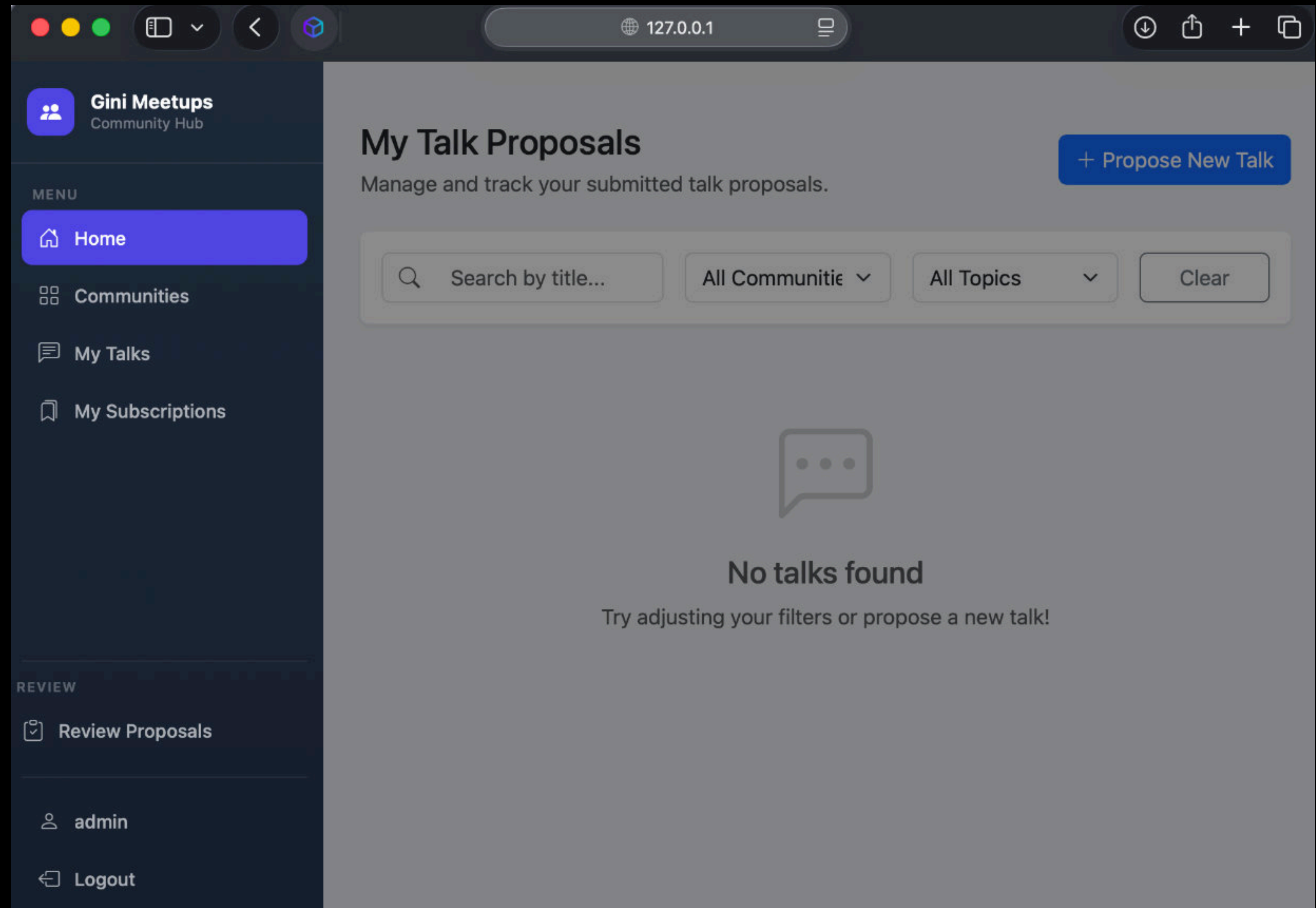
Imagination is your limit

- Advent of Code
- Games / graphics
- GUI desktop apps



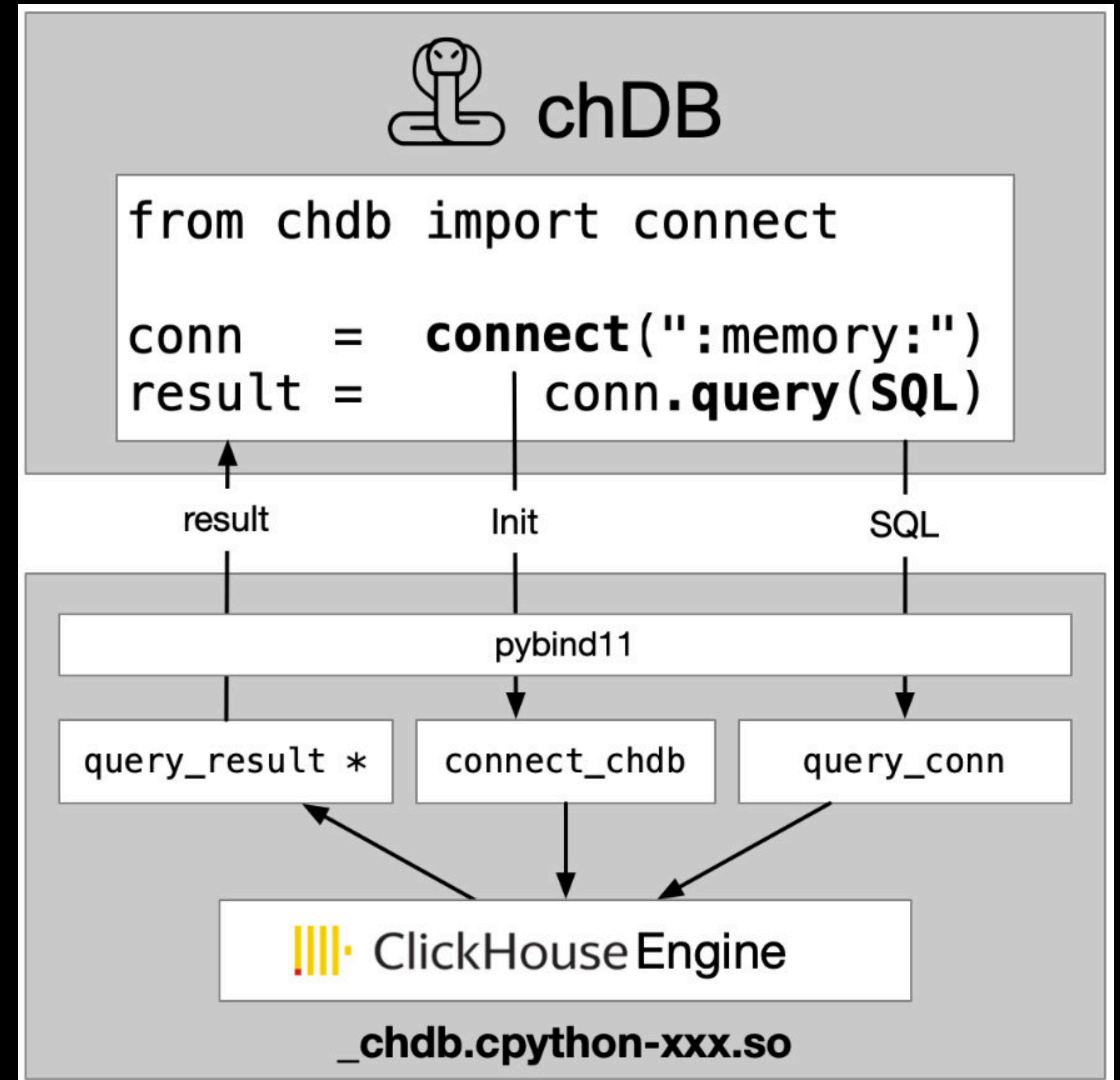
Imagination is your limit

- Advent of Code
- Games / graphics
- GUI desktop apps
- Web services



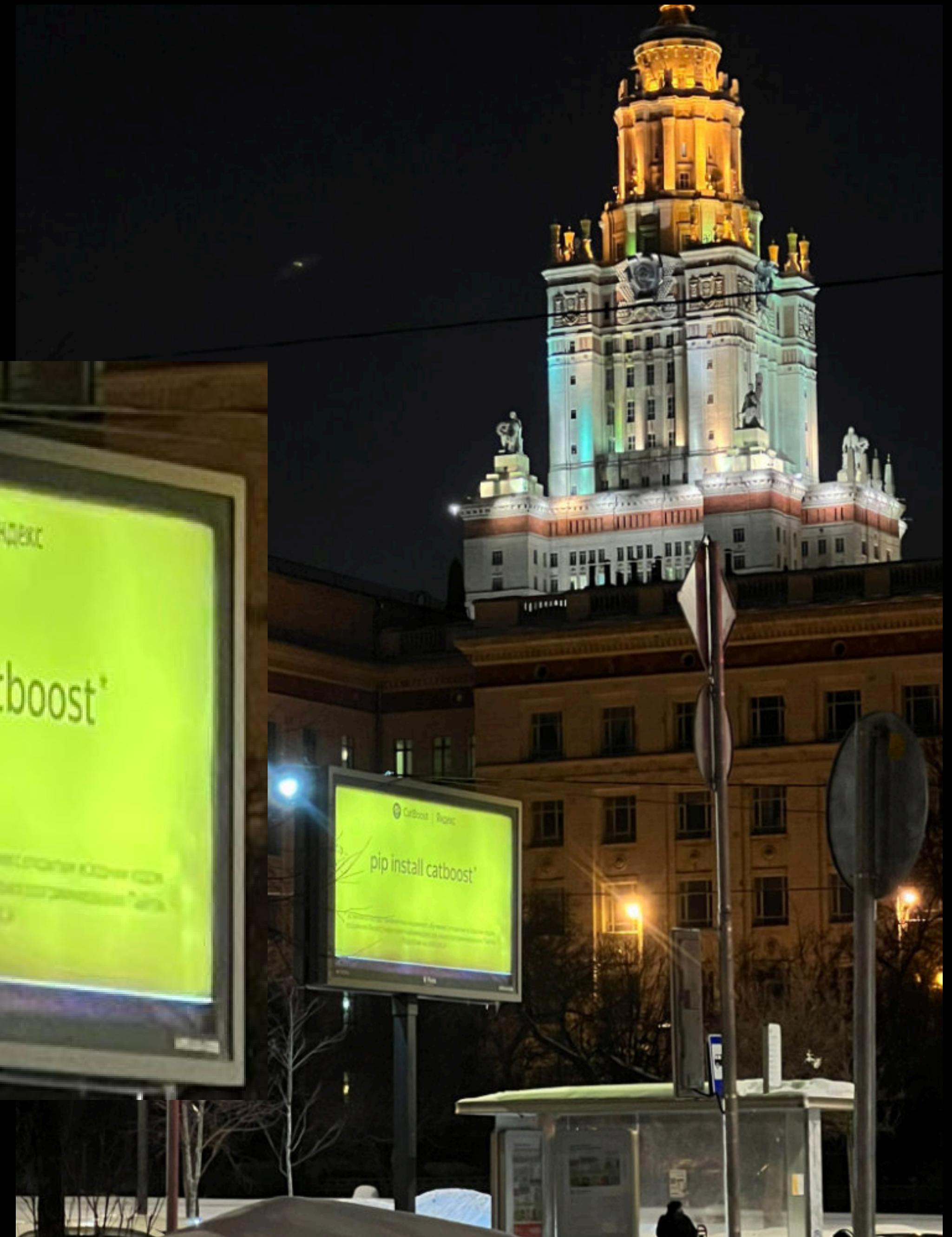
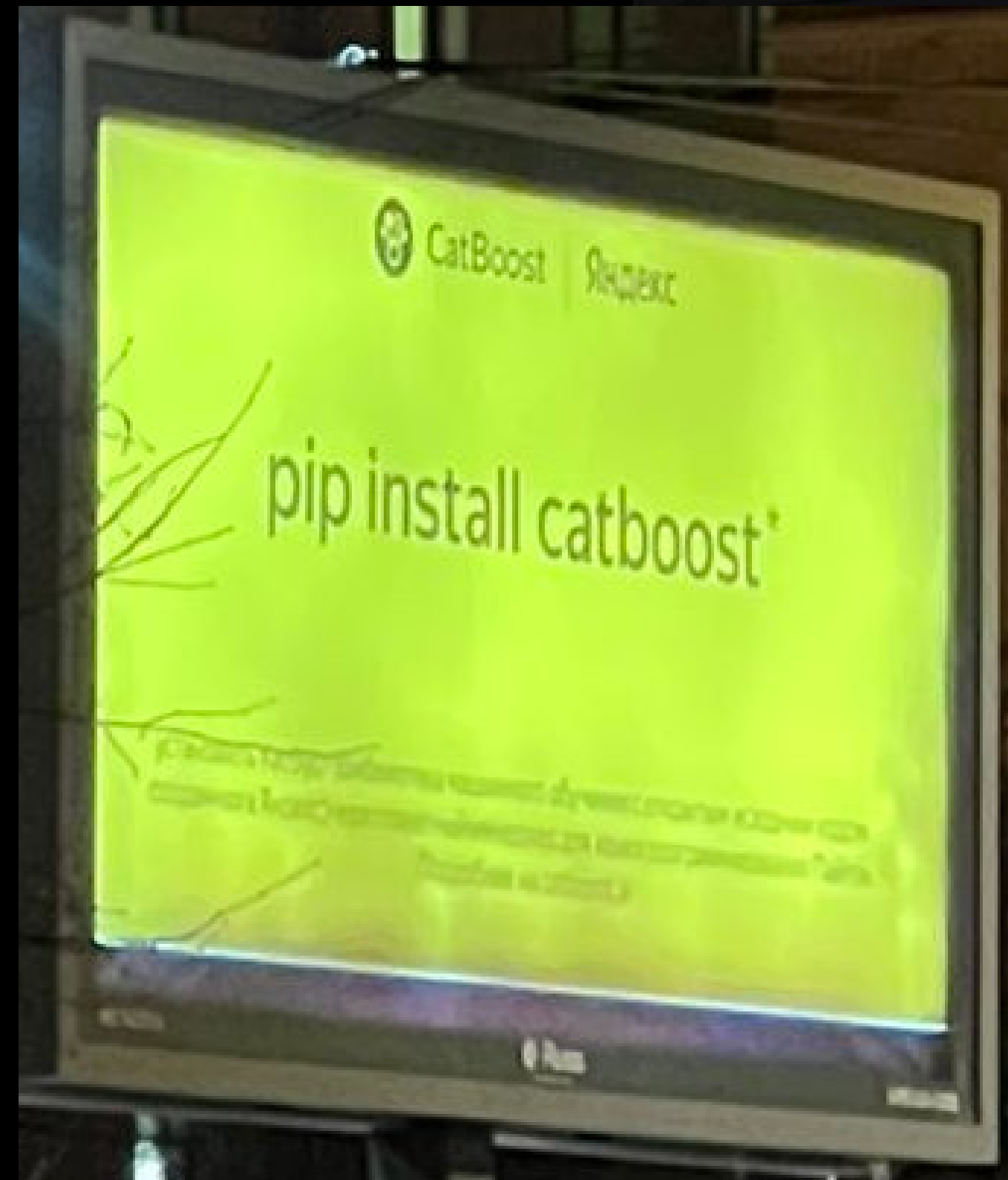
Imagination is your limit

- Advent of Code
- Games / graphics
- GUI desktop apps
- Web services
- ML libraries



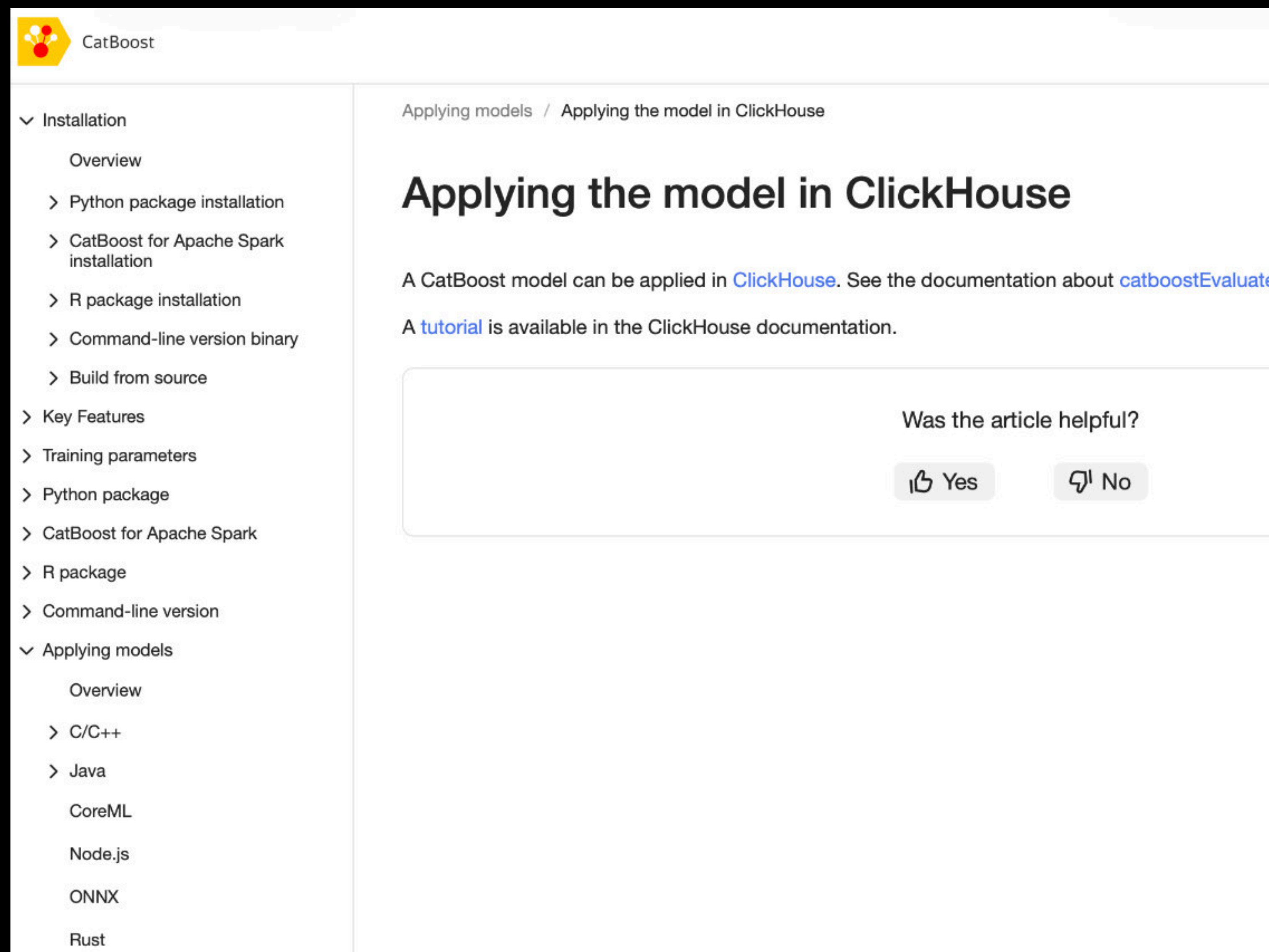
Imagination is your limit

- Advent of Code
- Games / graphics
- GUI desktop apps
- Web services
- ML libraries



Imagination is your limit

- Advent of Code
- Games / graphics
- GUI desktop apps
- Web services
- ML libraries



The screenshot shows the CatBoost documentation website. The left sidebar contains a navigation menu with sections: Installation (Overview, Python package installation, CatBoost for Apache Spark installation, R package installation, Command-line version binary, Build from source), Key Features, Training parameters, Python package, CatBoost for Apache Spark, R package, Command-line version, Applying models (Overview, C/C++, Java, CoreML, Node.js, ONNX, Rust), and a footer with '© 2018 Yandex'. The main content area is titled 'Applying the model in ClickHouse' and includes the text: 'A CatBoost model can be applied in [ClickHouse](#). See the documentation about [catboostEvaluate](#). A [tutorial](#) is available in the ClickHouse documentation.' At the bottom right, there is a feedback section titled 'Was the article helpful?' with 'Yes' and 'No' buttons.

Imagination is your limit

- Advent of Code
- Games / graphics
- GUI desktop apps
- Web services
- ML libraries

```
auto sess = &Session.sessionInstance();
sess.connect(":memory:");

auto res = sess.query(`
CREATE TABLE amazon_train
(
    date Date MATERIALIZED today(),
    ACTION UInt8,
    RESOURCE UInt32,
    MGR_ID UInt32,
    ROLE_ROLLUP_1 UInt32,
    ROLE_ROLLUP_2 UInt32,
    ROLE_DEPTNAME UInt32,
    ROLE_TITLE UInt32,
    ROLE_FAMILY_DESC UInt32,
    ROLE_FAMILY UInt32,
    ROLE_CODE UInt32
)
ENGINE = MergeTree ORDER BY tuple()
`, OutputFormat.CSVWithNames);
writeln("Creation table error: ", res.error);

res = sess.query(`INSERT INTO amazon_train FROM INFILE 'data/train.csv'`, OutputFormat.CSV);
writeln("Insert data error: ", res.error);
```

Imagination is your limit

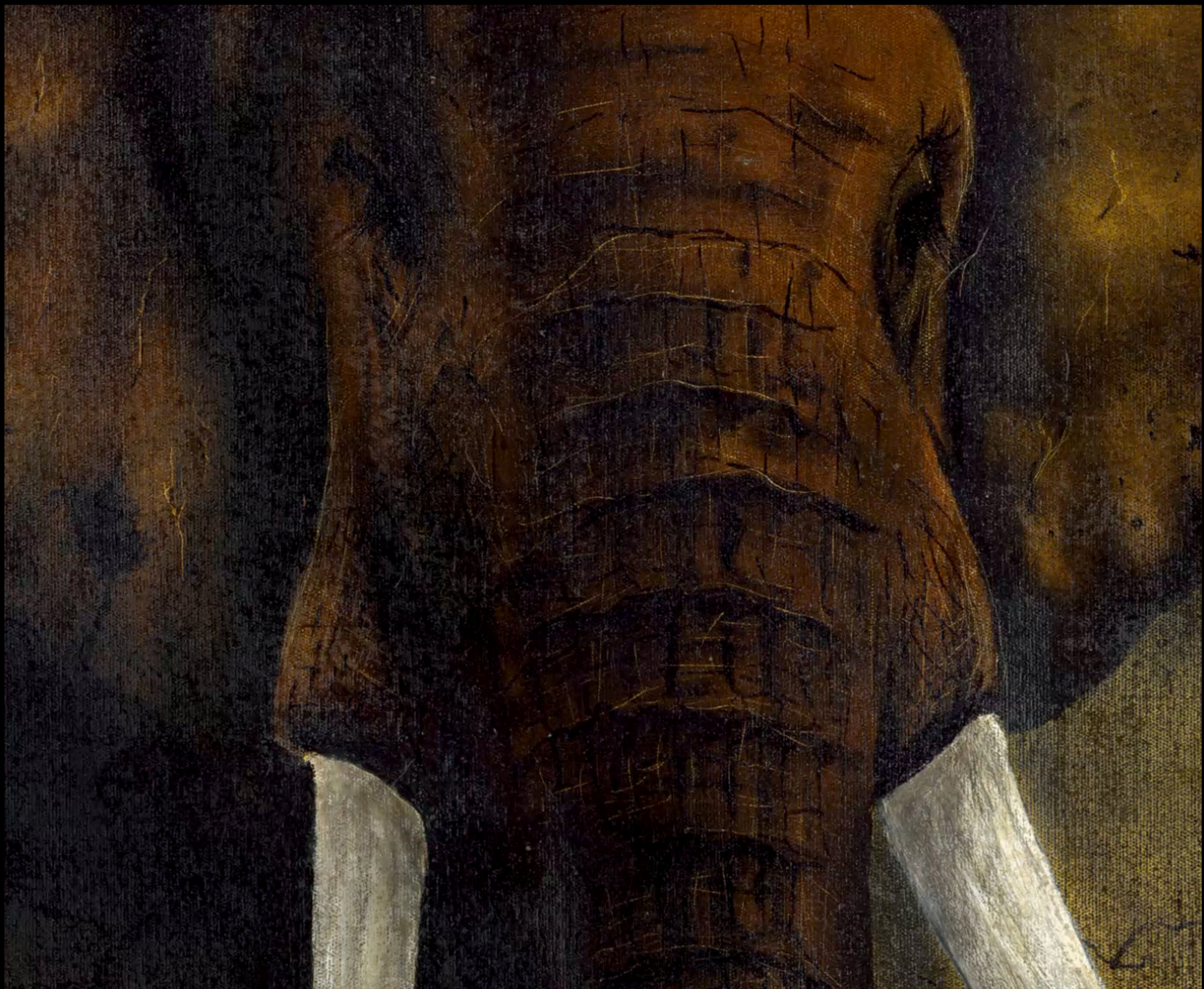
- Advent of Code
- Games / graphics
- GUI desktop apps
- Web services
- ML libraries

```
res = sess.query(`SELECT count() from amazon_train`, OutputFormat.CSV);
writeln("Select count result: ", res.buf);
writeln("Select count error: ", res.error);

res = sess.query(`
SELECT
    catboostEvaluate('cb_models/catboost_model.bin',
        RESOURCE,
        MGR_ID,
        ROLE_ROLLUP_1,
        ROLE_ROLLUP_2,
        ROLE_DEPTNAME,
        ROLE_TITLE,
        ROLE_FAMILY_DESC,
        ROLE_FAMILY,
        ROLE_CODE) > 0 AS prediction,
    ACTION AS target
FROM amazon_train
LIMIT 10`, OutputFormat.CSV);
writeln("—");
writeln("Data: ", res.buf);
writeln("Time elapsed: ", res.elapsed);
writeln("Select count error: ", res.error);
writeln("—");
```

One more thing





One more thing

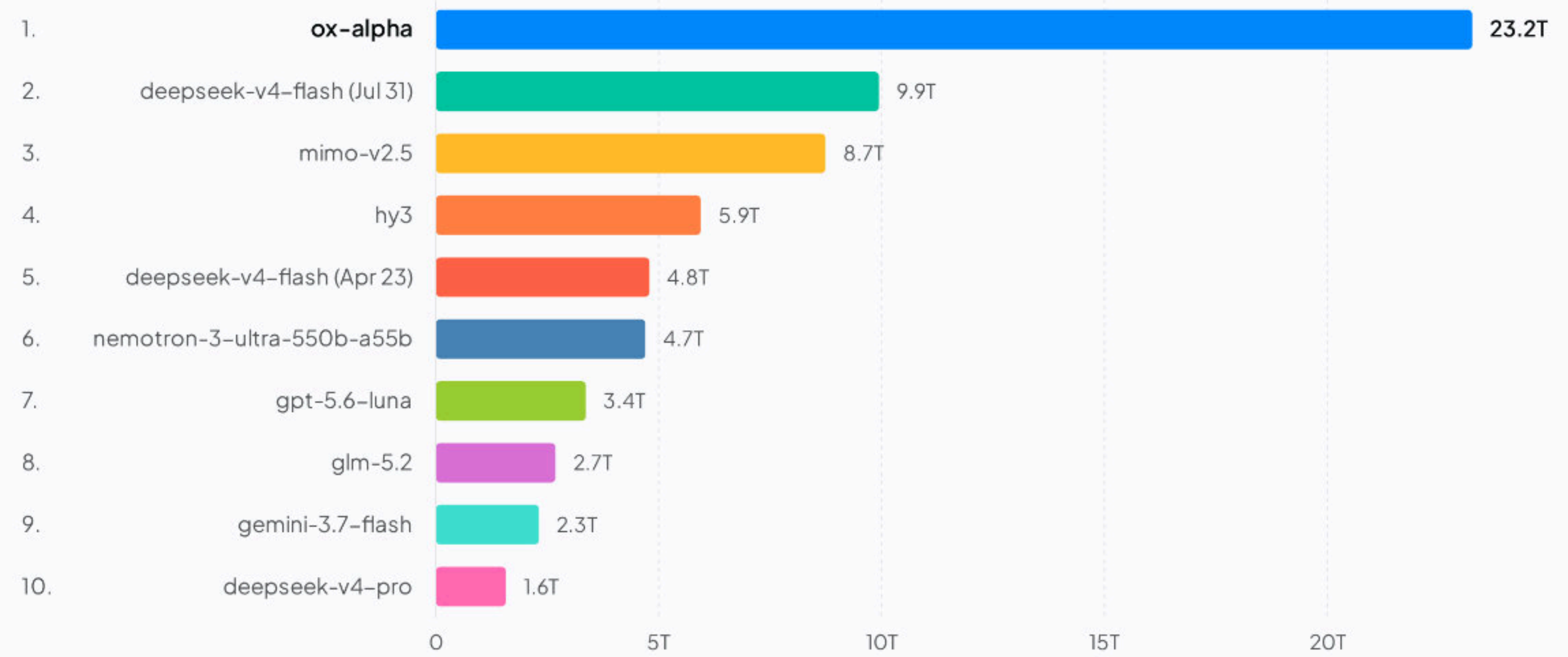
- GenAI
- Ox-alpha
- kdl-d
- symgc



OPENROUTER TRAFFIC · AUG 20-25, 2026 (UTC)

ox-alpha is the #1 model on OpenRouter

Tokens processed by model since ox-alpha launched (6 full days) · 2.3× the next model



One more thing

- GenAI
- Adoption of other **AFTER 7 YEARS IN PRODUCTION, SCARF HAS RELUCTANTLY MOVED AWAY FROM HASKELL**

Avi Press | July 10, 2026

- Haskell

DISCLAIMER

This has been a hard post to write. I almost didn't write it at all, since I prefer to build and promote than to critique. However, I hope this post can add constructively to the discussion about Haskell's future. I must underline that I'm not writing this criticizing Haskell from the outside. I care enough about Haskell to be honest about why Scarf has reluctantly moved away from it, in hopes it sways people in the community to take this feedback seriously.

One more thing

- GenAI
- Adoption of other languages
- Go

Why Go is an Ideal Language for AI-Assisted Software Engineering

AUG. 11, 2026

[Cameron Balahan](#)

Group Product Manager
Go

[Richard Seroter](#)

Chief Evangelist
Google Cloud

[Share](#)



One more thing

- GenAI
- Adoption of other languages
- Rust

ESSAY RECORD

Why Rust Wins in the Age of AI

RECORD

Essay / 2026.04.15 / 4 min read / 1,198 words

TAGS

[create](#) / [rust](#)

On this page

5 sections / +

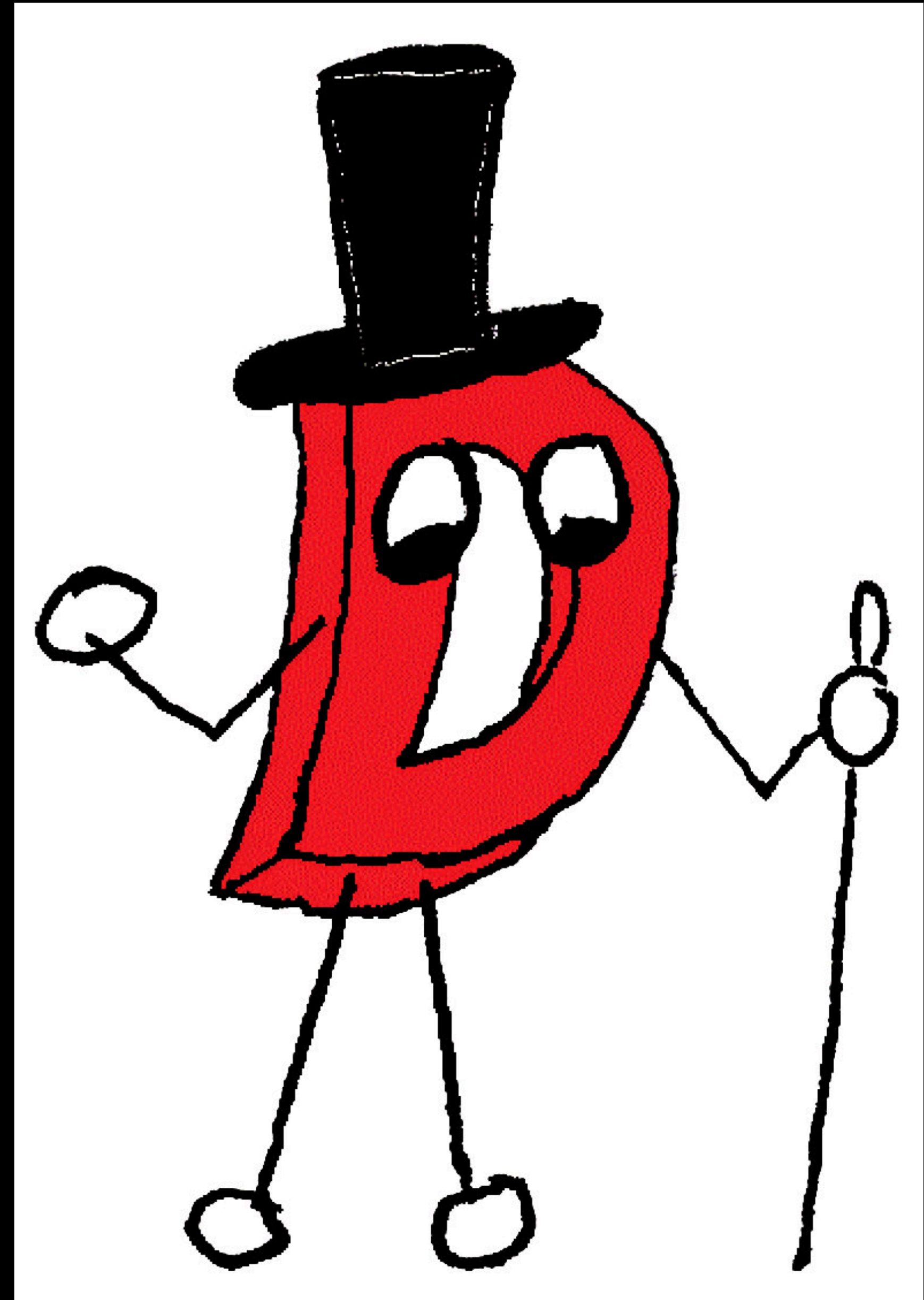
DISCLOSURE: Some links may be affiliate links. [Details](#)

For the past several months I've been searching for [the Missing Programming Language](#) - a language with a good balance of types, performance, ecosystem, and agentic AI performance. I've landed on a specific flavor of Rust - [High-Level Rust](#) - which gives me 80% of its benefits with 20% of its obstacles.

In this post I want to share why I think Rust is uniquely positioned for agentic engineering - providing a deterministic foundation upon which the nondeterministic models can build.

One more thing

- GenAI
- Adoption of other languages
- What about D?



Recap

NSA Suggested Memory-Safe Programming Languages

Controlled by Google

Controlled by Oracle

Controlled by FAANG

- Rust
- Go
- C#
- Java
- Swift
- JavaScript

Controlled by Amazon, Microsoft, Google, Huawei, Mozilla

Controlled by Microsoft

Controlled by Apple



uncontrollable

Discovery



Language and standard library



Ecosystem and community



Ports and bindings



Personal projects



With AI

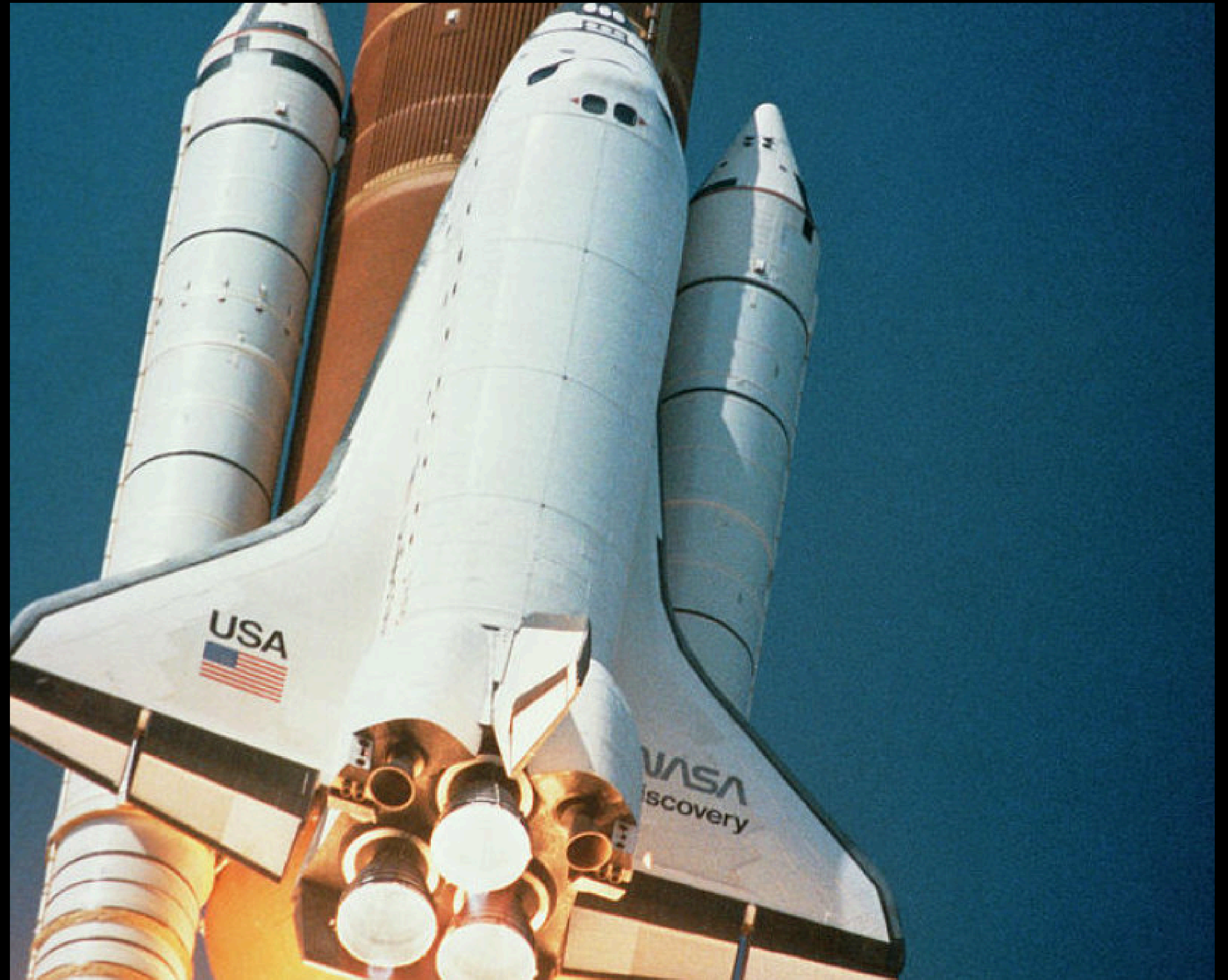


Recap for community

- Discovery

- Official resources

- dlang.org
- tour.dlang.org
- code.dlang.org



Recap for community

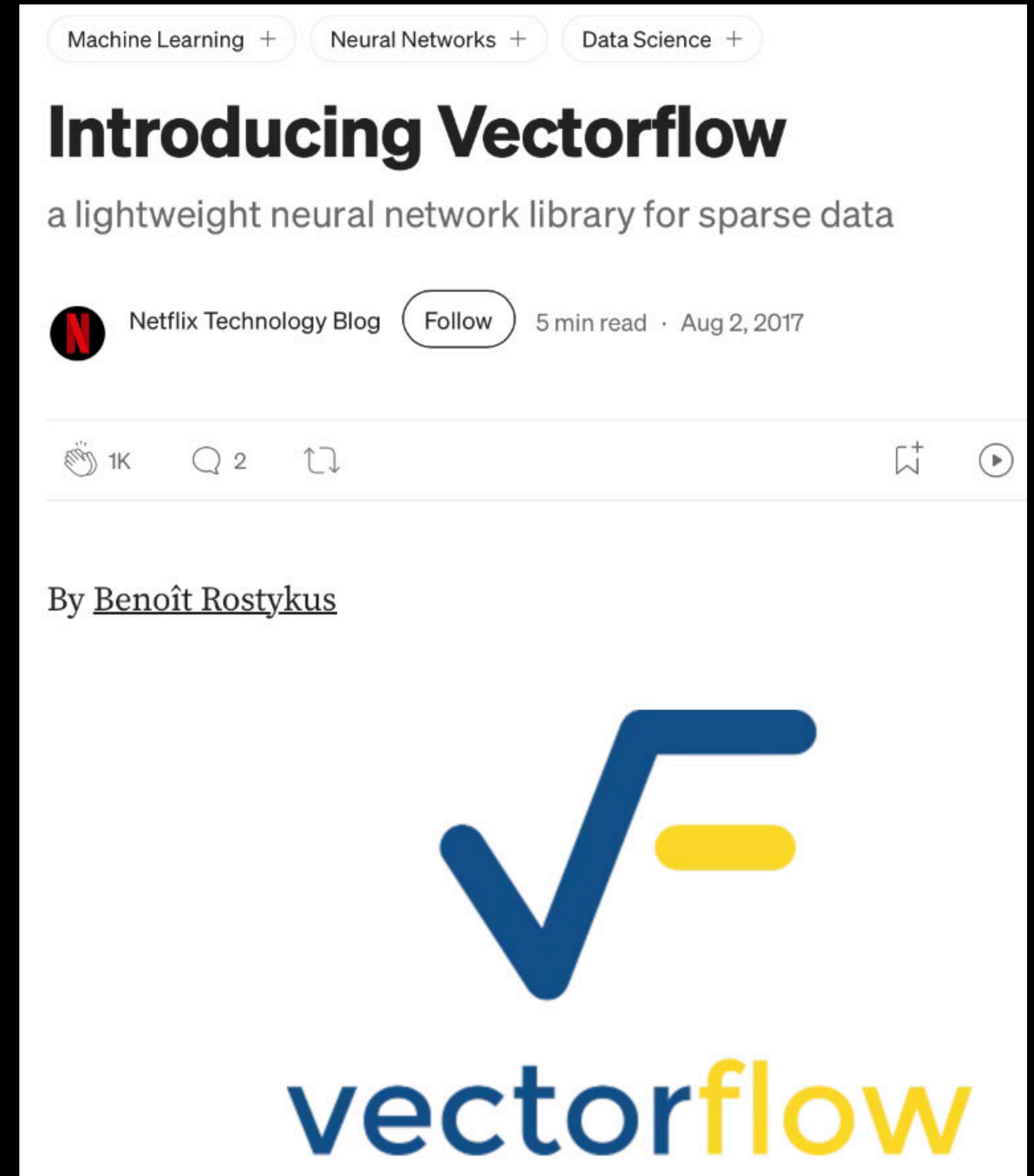
- Discovery
- Platforms

Website	Availability
Codeforces	 Yes
AtCoder	 Yes
CodeChef	 No
LeetCode	 No
CSES	 No
HackerRank	 Yes
Kattis	 Yes
Topcoder	 No
DMOJ	 Yes

Recap for community

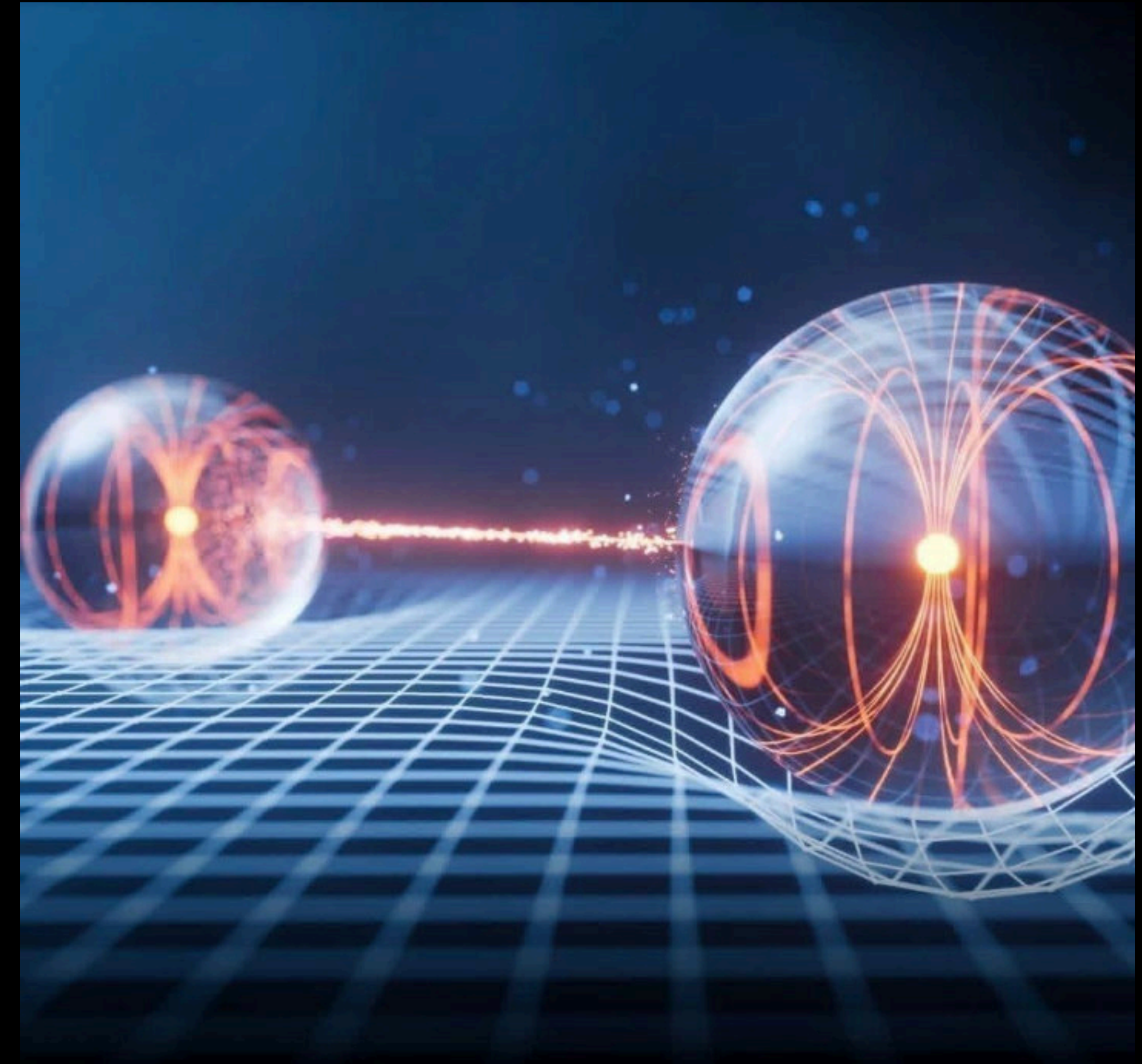
- Discovery

- Be loud



Recap for community

- Discovery
- Momentum



**Photons Hold No Mass
But Have Momentum**

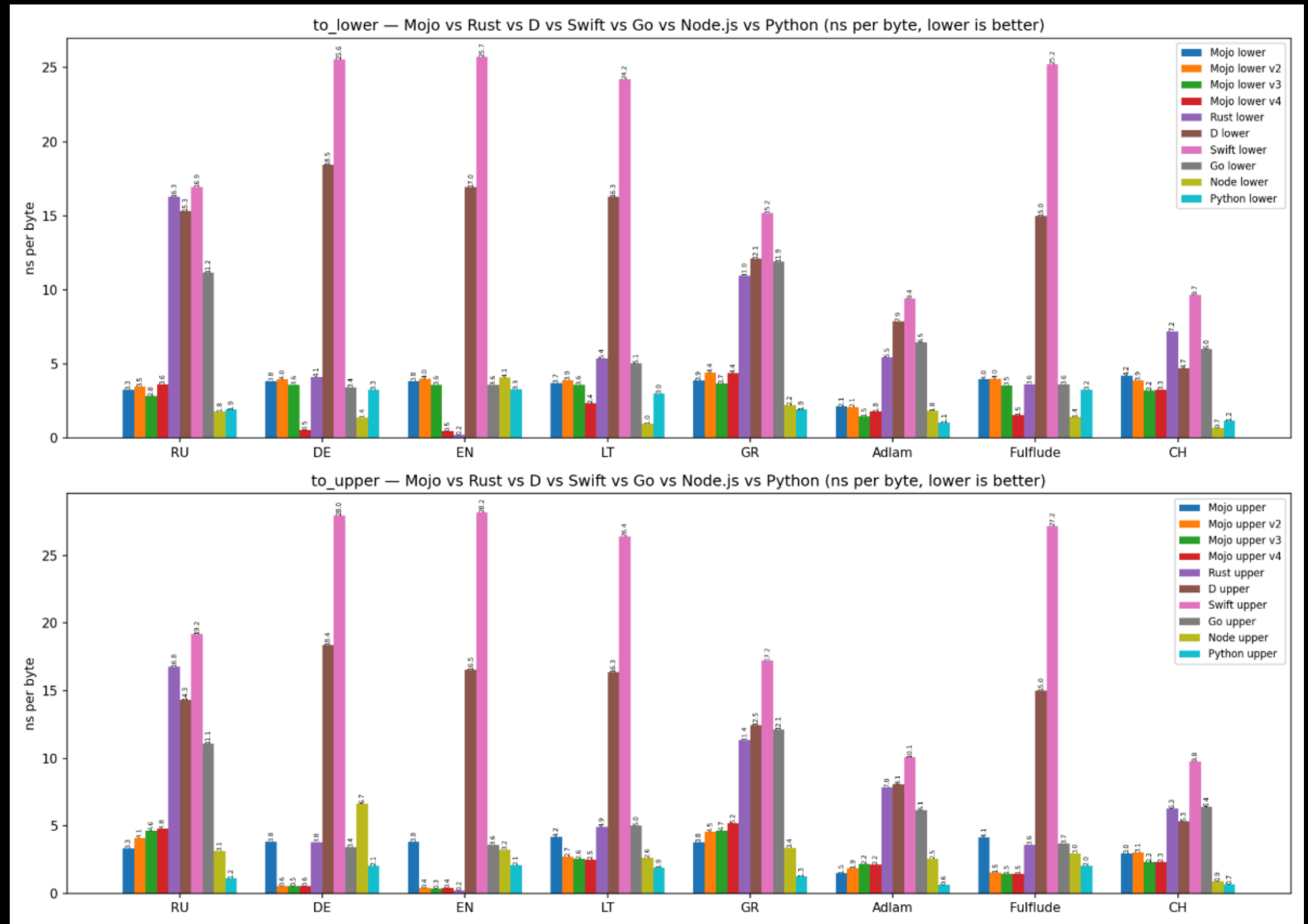
Recap for community

- Discovery
- Momentum
- Curiosity



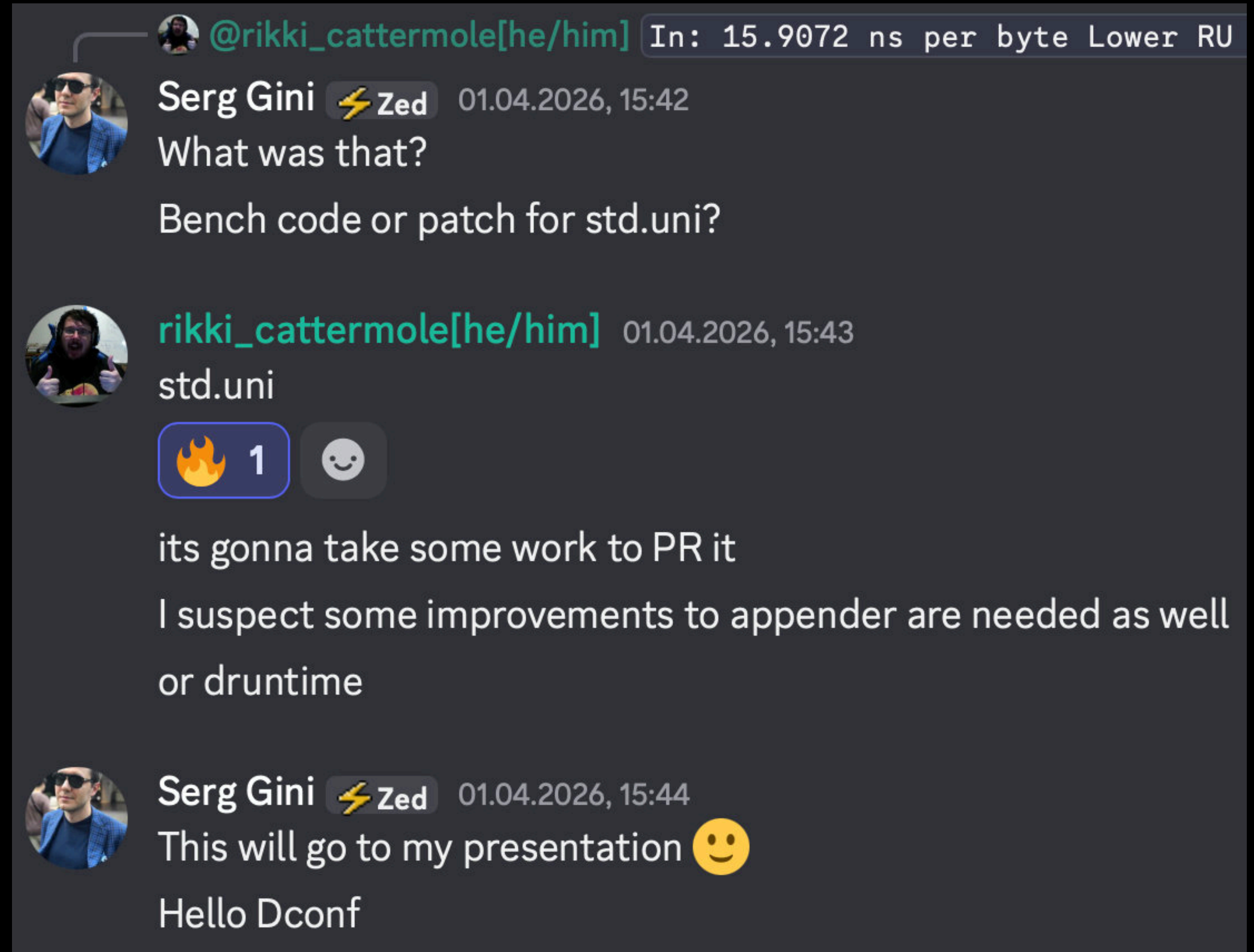
Recap for community

- Discovery
- Momentum
- Curiosity



Recap for community

- Discovery
- Momentum
- Curiosity



Recap for community

- Discovery
- Momentum
- Curiosity

Improve std.uni:toCase performance #10986

 **Merged** [thewilsonator](#) merged 1 commit into [dlang:master](#) from [rikkimax:slow-toCase](#)  on Apr 3

 Conversation **5**

 Commits **1**

 Checks **7**

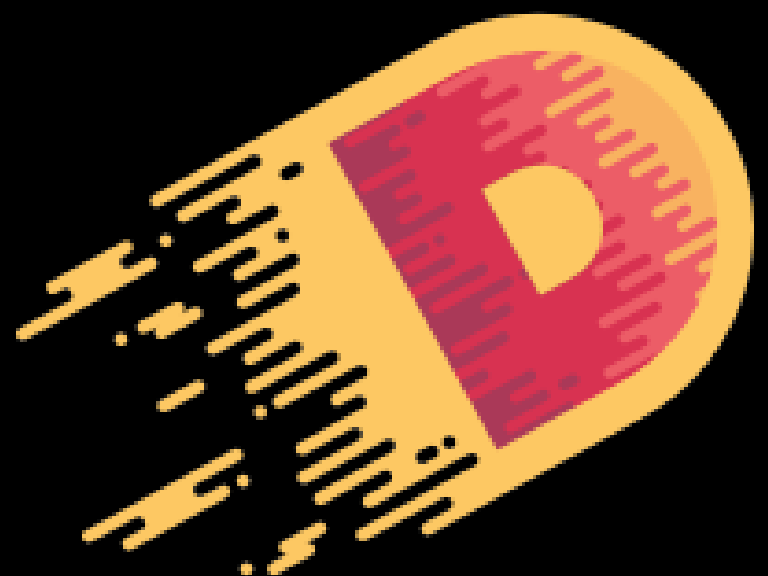
 Files changed **1**



rikkimax commented [on Apr 2](#)

For the benchmark that Serg pointed me at as being slow, it went from 26ns down to 7ns.





Thank you

Talk materials



done with Chitra

<https://github.com/cyrusmsk/talks/tree/main/dconf2026>